Grandfathering at Sale: Nonportable School Rights and Housing Exchange

Tai Lo Yeung*
Università della Svizzera italiana

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Abstract

I study housing exchange when a public entitlement survives retention but expires at sale. Beijing's 2018 reform preserved single-school assignment for existing titles but placed buyers under multi-school assignment after transfer. Across fifty-nine communities, formerly key-school communities remained near six completed sales while matched ordinary communities rose from three to five. The 2019 formerly-key/ordinary after-before ratio is 0.42–0.46 times its 2018 counterpart; older and averaged benchmarks preserve the sign but imply smaller gaps. Within listings observed to complete, posting changes little; selected-stock and early-completion point estimates suggest slower execution, while active-stock points lie closer to one than completed flow. Among completed sales, marketing-time point estimates rise while prices and discounts move about one percent. No benchmarked two-year difference is detectable. A search-and-bargaining model shows how the title reset raises the match threshold. In partial equilibrium, attaching protection to the household or property expands the match set.

*Email: tai.lo.yeung@usi.ch. I thank Lorenz Kueng, Francisco Amaral, Alberto Plazzi, and Jessica Wachter for helpful comments and suggestions. All errors are my own.

1 Introduction

Grandfathering solves one problem and can create another. It protects people who relied on an old rule, but protection that survives only while an asset is retained makes exchange the forfeiture event. The stakes can be large: using a representative nationwide dataset, Batzer et al. (2026) estimate that mortgage-rate lock-in was associated with 1.72 million fewer arm's-length transactions between 2022Q2 and 2024Q2. California's property-tax portability reforms show the reverse margin, raising mobility among eligible older homeowners (Ferreira, 2010). The same logic applies whenever a favorable mortgage, property-tax basis, license, or public benefit is tied to the incumbent rather than transferable with the asset (Fonseca and Liu, 2024). I ask what happens when a public right disappears precisely because its owner trades—and what object a transition right should follow.

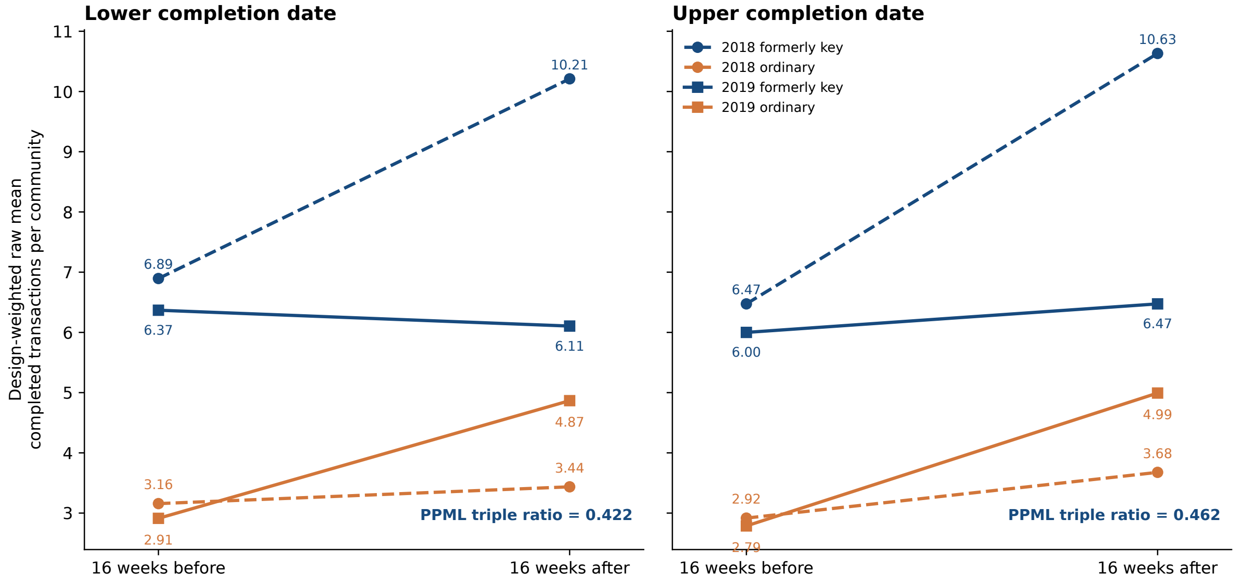
The main takeaway is simple: *when sale destroys a grandfathered right, adjustment runs through fewer and slower completed trades, while the prices of the trades that clear move little.* If an amenity changes symmetrically for an owner and every prospective buyer, its value can be capitalized into price. Here the incumbent and buyer value different bundles. The owner keeps one entitlement by retaining the asset; after transfer, the buyer receives another. Price can divide the gains from trade that remain, but it cannot recreate the entitlement extinguished by sale. When the old right is more valuable than the expected post-transfer bundle, sale raises the bilateral match threshold.

Beijing's Haidian district provides a sharp setting. Residential addresses historically mapped eligible households to particular public primary schools. A reform announced in 2018 preserved the prior single-school rule for titles registered before January 1, 2019 while they remained under the old title. A buyer whose acquisition generated a post-cutoff title instead entered multi-school assignment (Haidian District Education Commission, 2018a,b). The dwelling did not change at transfer; the school-assignment regime obtained through purchase did. The policy supplies a legal date, a pre-policy map of differential exposure, and a direct reason why the seller's retained bundle differs from the buyer's acquired bundle.

The central empirical fact is easy to state: *six stays six while three becomes five*. During the sixteen weeks around January 1, 2019, formerly key-school communities remain near six completed transactions per community, while matched ordinary-school communities rise from roughly three to five. Figure 1 shows this pattern under both feasible completion-date conventions and the identical 2018 calendar comparison used to benchmark differential winter-to-spring seasonality.

The policy changed both groups, so ordinary-school communities are not an untreated control. The design instead measures differential policy incidence: replacing a prior key-school mapping is more consequential than replacing an ordinary-school mapping. A seasonal triple difference compares the formerly-key–ordinary winter-to-spring differential in 2019 with the same calendar differential in 2018. The preferred frame contains fifty-nine notice-verified communities, nineteen historically assigned to seven schools classified as key, 7,340 completed transactions, and 1,173 transactions in the four principal cells.

Figure 1. Completed transactions around January 1 in formerly key and matched ordinary communities



Notes: Each point is an inverse-reuse-weighted raw mean of completed transactions per community in a sixteen-week window. Color identifies formerly key-school and ordinary-school communities; dashed and solid lines identify the 2018 identical-calendar comparison and the 2019 policy experiment. The PPML triple ratio compares the 2019 formerly-key/ordinary after-before ratio with its 2018 analogue; it is not a districtwide volume ratio.

I establish the argument in four steps. First, I document that formerly key-school communities do not join the 2019 winter-to-spring expansion observed in Beijing outside Haidian, other Haidian communities, and the matched ordinary group. The designated seasonal comparison yields a completed-flow ratio of 0.42–0.46. Older and averaged benchmarks retain the sign but imply smaller gaps, making the direction more stable than the magnitude. In an exhaustive fixed-weight relabeling diagnostic, the actual grouping produces a more negative coefficient than all but 275 and 1,742 of the 116,280 seven-school labelings at the two date conventions.

Second, I follow the gap through the exchange process. Within listings observed to complete by the archive end, posting incidence is near one at the point estimate while selected-stock and early-completion point estimates shift in the direction of slower execution. Independent active-stock points also lie closer to one than completed flow. Among completed sales, marketing-duration point estimates rise by 0.70–0.75 log points while transaction-price and discount point estimates move by about one percent. A floor-area split recorded before its estimation is more negative for larger units under the designated 2018 benchmark, but the contrast attenuates under older benchmarks and the bedroom split is estimator-sensitive. A separately benchmarked two-year comparison detects no cumulative difference. These measures use different risk sets; jointly they describe a configuration consistent with slower selected execution, not a marketwide listing hazard or a uniquely identified mechanism.

Third, I develop a search-and-bargaining model in which the title reset raises the bilateral match threshold. Conditional on match opportunities, a higher threshold removes marginal matches, slows clearance, changes the location composition of accepted flow, and changes selection into observed prices. The completed-flow response is signed; the conditional-price response is not. This explains why a large relative-flow gap and small selected-price movement can coexist.

Fourth, I use the same primitives to compare four transition rules. Protection can follow the standing title, the property, the household, or a common calendar date. When the prior mapping is at least as valuable as the expected post-transfer bundle, property portability, household portability, and a uniform transition each weakly expand the bilateral match set relative to title-contingent grandfathering. This is a partial-equilibrium match-set ordering, not a welfare ranking.

The analysis keeps three observation universes separate. A 586,312-transaction platform archive measures completed exits; posting and completion clocks provide exact accounting within the selected set observed to complete by the archive end; and independent public crawler snapshots measure active offers on fixed supports. Historical admission notices reconstruct pre-policy school assignments without using transaction outcomes, and a fixed one-day interval bounds missing completion dates. I probe the result with few-cluster inference, assignment restrictions, alternative windows, leave-outs, and outcome-blind spatial matching; the online appendix contains the complete audit and protocol history.

Related literature. I make three contributions. First, school-capitalization studies examine how access enters property values (Black, 1999; Gibbons and Machin, 2003; Bayer et al., 2007; Fack and Grenet, 2010), while recent Beijing work studies price, sorting, and spatial responses to multi-school zoning (Guan et al., 2024; Yang et al., 2024; Dai et al., 2025, 2026; Tong et al., 2024; Chen et al., 2026). The object here is different: a sale-triggered change in the bundle being exchanged rather than a common neighborhood amenity shock. The closest published volume evidence is Shao et al. (2023), who study announcement and enforcement responses after a later Xicheng reform.[1]

Second, the paper extends evidence that nonportable incumbent benefits suppress mobility—including property-tax grandfathering and mortgage-rate lock-in (Ferreira, 2010; Fonseca and Liu, 2024; Batzer et al., 2026)—to a public right bundled with housing. Following completed flow, the observed-completer pipeline, active-stock snapshots, clearance, and selected prices separately makes the market-stage contribution. Third, the model treats the attachment rule itself as a policy choice and shows that reliance protection and protection conditional on non-sale are conceptually distinct.

The evidence supports a deliberately bounded conclusion. The headline estimate is a benchmark-sensitive, short-run relative exchange gap, not an absolute market collapse. The distinct observation universes do not identify full-market entry, withdrawal, or a sale hazard,

[1] Zhang (2022) describes a completed Beijing school-allocation project keyed to transaction timing, but no public manuscript or findings were located as of August 2026.

and the data do not identify a structural entitlement value or welfare ranking. Those limits discipline the mechanism without erasing the central fact that completed exchange responds much more than the prices observed among trades that clear.

The remainder of the paper presents the institution, data, and design; establishes the completed-flow fact; develops the model; tests its implications across the exchange funnel and longer horizons; and compares transition rules.

2 The Policy: Grandfathering Ends at Sale

2.1 School access as a housing entitlement

Public primary-school access in Beijing has historically been tied closely to household registration and residence. In the single-school regime, an eligible address was mapped to one responsibility school. That mapping gave ownership an entitlement-like component: a household buying a home in the zone could use the address when applying for admission, subject to household-registration and occupancy requirements. A large empirical literature finds that school access capitalizes into housing values because residence and enrollment are jointly chosen (Black, 1999; Bayer et al., 2007; Fack and Grenet, 2010). In Beijing the institutional link is particularly visible in the category of "school-district housing."

The relevant treatment is not simply whether a home was located in Haidian. Every community in the estimation sample was in the district, and every one faced the reform. Former key-school status proxies for the value of the prior single-school mapping that was lost; it is not an observed household-specific treatment intensity. I classify schools using the 67-school union reported in Appendix Table A1 of Zheng et al. (2025), which combines a 2013 China Study Center ranking with a widely circulated online list; it is not an official government designation. A community enters the treated group only when a visibly dated 2017 enrollment-notice image names it in the school's admission scope. Campus-brand, renamed-school, and loose affiliation links are excluded. Ordinary controls come from the pre-reform platform school map, exclude every strict or affiliated key-school match, and are chosen by geography within Haidian.

2.2 The title-vintage rule and its implementation

The policy document was dated April 28, 2018 and published April 30. In the author's translation, its operative clause states: "Beginning January 1, 2019, houses newly registered and issued a real-property certificate will no longer correspond to one school; enrollment will follow multi-school zoning" (Haidian District Education Commission, 2018a). A December 2018 district notice repeated the same effective date and rule (Haidian District Education Commission, 2018b). The 2018 notice does not publish each address's post-reform school set, assignment probability, or capacity constraint. Hence the policy created a known title-vintage cutoff but not an observed continuous treatment dose.

Figure 2. School-assignment rules before and after a title transfer

Notes: Keeping a pre-cutoff title preserves the prior single-school rule. Selling generates a new title, so the buyer enters multi-school assignment. The physical dwelling transfers; the incumbent's assignment rule does not. Admission remains subject to household-registration and occupancy requirements.

An official district retrospective reports that computerized allocation for families who bought after the title cutoff was first used in 2021 (Haidian District, 2022). The January 2019 design therefore identifies market responses to the title rule and its expected assignment consequences, not responses to contemporaneous realized lottery draws.

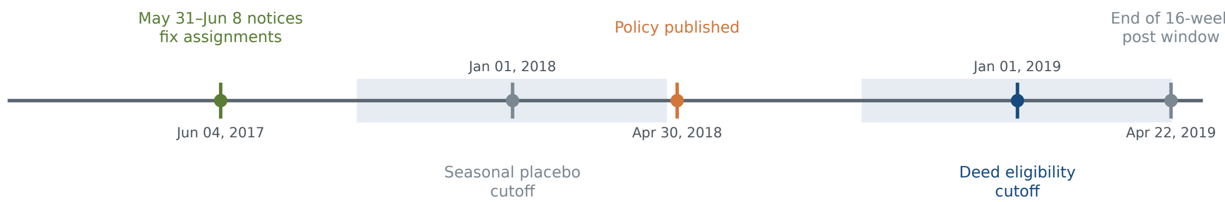
Later implementation materials make the post-transfer uncertainty more concrete without revealing a transaction-specific treatment dose. For the first post-cutoff cohort allocated by computer in 2021, the registration school was the first choice and guardians selected the remaining options. Applicants were admitted directly when demand did not exceed the allocation plan and were allocated by computer when it did. The number of options displayed depended on remaining places in the local area and could differ across families. Post-transfer access was therefore capacity-contingent, but the public materials do not report school-level plans, applicant counts, family-specific menus, or assignment probabilities for the transactions in the estimation window (Haidian District Education Commission, 2021; Beijing Evening News, 2021).

The title-vintage clause is economically central. A pre-cutoff title remained under the prior single-school rule while held; a buyer whose acquisition generated a post-cutoff title entered the new assignment regime. Admission still depended on household-registration and occupancy requirements. The reform therefore did not simply lower a neighborhood amenity for everyone at once; it made the governing school-assignment rule conditional on whether the asset was transferred.

Figure 2 makes the economic contrast explicit. If the incumbent keeps the dwelling, the physical house and the prior single-school mapping remain bundled. If the dwelling is sold, the physical asset passes to the buyer but the assignment regime resets. Price can divide the surplus from the house; it cannot make the buyer's post-transfer mapping identical to the incumbent's retained mapping. The empirical exposure measure asks where that reset was likely to matter most.

The economic prediction is ambiguous outside a narrow model. Before the cutoff, households could accelerate eligible transfers to preserve the prior single-school mapping.

Figure 3. Policy and identification timeline



Notes: The shaded regions are the sixteen-week estimation windows around January 1 in 2018 and 2019. January 1, 2018 supplies the identical-calendar seasonal comparison and is not a policy cutoff.

After the cutoff, the expected school bundle could fall, assignment uncertainty could rise, and buyers and sellers could take longer to locate an acceptable surplus. A broad grace-period surge is therefore one possible response, but not a logical necessity. The analysis treats that announcement-period response and the local enforcement response as distinct predictions; both were specified before reconstruction of the missing completion dates.

The cutoff is legally defined by registration and certification, not by the platform's transaction-completion timestamp. Administrative processing may separate the two. On March 1, 2019, Haidian issued its first certificate under Beijing's newly introduced integrated transaction–tax–title-registration service; the official account states that total administrative handling was compressed to four working days (Beijing Municipal Commission of Planning and Natural Resources, 2019). The source does not establish that the platform timestamp is the certificate-issuance date. I therefore treat this concurrent change as a timing threat, not as evidence that the school rule caused the broader winter-to-spring rise. A districtwide process change can generate the triple difference only if its take-up or transmission into the platform clock differs systematically by former school status. These timing distinctions are why I use sixteen-week windows, show bandwidth and donut sensitivity, and avoid the language of unit-level sharp treatment assignment. The estimand is a differential completed-flow response around enforcement, not a local average treatment effect for homes one day apart in deed issuance.

3 Data: Three Views of Housing Exchange

The observed outcome is housing exchange, not household migration: the archive does not link sellers or buyers across addresses.

The project combines three empirically distinct objects. First, the canonical platform archive contains completed resale transactions and supplies the main outcome. Within that same archive, I use formerly key, matched ordinary, other Haidian, and Beijing-outside-Haidian observations as four mutually exclusive descriptive benchmark groups for the observed policy-year winter-to-spring rise; they are neither independent research designs nor untreated policy controls. Second, posting and completion clocks within the archive construct a selected pipeline consisting only of listings observed to complete by the archive end. Third, independent public crawler snapshots observe active offers at particular dates. Keeping these objects separate is not semantic housekeeping: it determines which denominators exist and which market stages can be studied.

3.1 Completions, successful-listing spells, and active stock

Completed transactions. The main archive records successful exits from the resale process. A month-only source completion date is still a completed transaction when the row carries a positive deal price and a marketing-duration clock. Missing a parsed day is therefore not evidence of non-sale. These records identify the timing and location of platform-recorded completions, but not the number of properties ever offered.

The observed-completer pipeline. For completed records, posting dates reveal when each successful listing entered the platform pipeline. This creates exact posting, stock, and exit accounting within the selected set $I_i = 1$ of listings recorded as complete before the archive ends in January 2021. Both January–April posting cohorts used below receive at least 540 days of common follow-up, but conditioning on $I_i = 1$ still excludes withdrawals, unresolved listings, and later completions by construction.

Public active-stock snapshots. Three independently collected public files contain genuine active listings: a November 2017 crawl with exact listing IDs (Huang, 2017), a late-March 2018 cross-section distributed with the H4M data (Zhao et al., 2022), and an archived project crawl collected community-by-community on February 27, 2019. These files establish that listing observations exist outside the completed archive. They do not share stable listing identifiers, timestamps, or crawler ledgers and therefore cannot be stacked into a property-level transition panel. I use them only on identical community supports and expose the cross-crawler comparability assumption through a sensitivity parameter.[2]

3.2 Transaction archive and estimation sample

The transaction archive is a processed export from a proprietary listing platform covering Beijing resale records posted from 2010 through January 2021; the analysis period is 2015–2020. The canonical input contains 586,312 completed transactions after the project's cleaning screens and supplies both the four geographic archive benchmarks and the estimation frame. Every retained record has a positive transaction price, posting date, and marketing duration. Some observations lack a parsed day-level completion date because the

[2]The provider's auxiliary listing/attention export also includes rows without published transaction prices. In the fifty-nine-community frame, only nineteen such rows have 2018 posting years and sixty-one have 2019 posting years; their observed status dates begin on April 23, 2019 and are asynchronous. They confirm that non-completer records exist but do not form a January 2019 risk set.

source date is month-only, but that field does not indicate that the property failed to sell. The analysis therefore uses counts of platform-recorded completed transactions rather than a listing-level sale probability.

For the preferred notice-verified Haidian frame, the archive supplies 7,340 resolved transactions across fifty-nine communities: nineteen exposed communities and forty ordinary-school comparisons. Of these, 1,173 fall inside the four sixteen-week estimation windows. The regression contains 236 community–experiment–side observations. The online appendix summarizes the estimation samples and reports the archive-wide measurement audit.

3.3 Completion-date measurement

Let a_i be listing i's posting date and W_i its recorded marketing duration. When an exact completion date d_i is present, I retain it. When it is missing, I use the fixed interval

$$d_i \in [a_i + W_i - 1,\ a_i + W_i]. \tag{1}$$

The lower endpoint follows the modal inclusive-day convention; the upper endpoint is the one-day sensitivity. The convention was fixed before re-estimating the policy coefficient.

The validation sample contains 414,979 transactions with both an exact date and the duration clock. Reconstructed and exact duration have correlation 0.99999, and all but one reconstructed date lies within one day of the reported date. In the broader sixty-one-community matched frame, 2,515 of 7,410 dates use the reconstruction; twenty-seven of those reconstructed dates (1.1 percent, or 0.36 percent of all rows) change cutoff-cell membership between endpoints. Both endpoints remain visible throughout the completed-flow analysis. The archive does not document platform market share, off-platform trades, or a stable coverage denominator; a treatment-correlated coverage change therefore remains possible.

3.4 Historical exposure and comparison communities

Historical exposure is fixed from 2017 school-admission notices. The exposed sample contains nineteen communities tied to seven schools externally classified as key. An outcome-blind archival reconstruction of the 2018 notice images verifies that all nineteen exposed communities remained assigned to the same seven schools one year later; this check neither selects units nor rematches the design. The images are retained with source pages and hashes, but come from a public secondary archive rather than an authenticated district database. Among the original ordinary-school comparisons, thirty-one communities appear literally in the 2017 notice text and nine are linked by an exact, unambiguous property or address alias. The preferred sample retains those forty controls; an exact-notice sensitivity retains only the thirty-one literal matches. Neither restriction rematches the treated communities.

Each exposed community is matched to as many as five nearby eligible ordinary-school communities, prioritizing the same pre-reform school district and otherwise remaining within

Haidian. Reused controls receive inverse-reuse weights, normalized within the matched design. Inference clusters by historically assigned school. The preferred sample contains twenty-one school clusters, seven of which are exposed, so the headline analysis exhausts all 2^{21} assigned-school Rademacher sign vectors under the restricted-null, fully studentized wild procedure and also reports leverage-aware CR2/Satterthwaite inference (Pustejovsky and Tipton, 2018) together with leave-one-out diagnostics. This enumeration removes Monte Carlo error but is not assignment-based randomization inference.

The external key-school classification is not created from the policy outcome. As an ex ante validation, I estimate a school component from 2015–2016 transaction prices and ask whether it predicts 2017 values before the announcement. The component lowers 2017 transaction-level mean squared error, its school-mean validation slope is 0.934, and formerly key schools score 0.851 standard deviations above ordinary schools. The online appendix reports the full diagnostic and explains why the externally classified binary ordering remains the primary exposure measure.

3.5 Outcome construction and aggregation

The main dependent variable is the number of platform-recorded completed transactions in each community–experiment–side cell. Each community appears four times: before and after January 1 in the 2018 seasonal experiment and before and after January 1 in the 2019 policy experiment. Zero-count cells are retained. The preferred linear specification applies the inverse hyperbolic sine to accommodate zeros without adding an arbitrary constant; PPML estimates the same saturated contrast on the count scale.

The sixteen-week window is long enough to accommodate search and the administrative separation between platform completion and title certification while remaining local to enforcement. I report the complete four- to sixteen-week bandwidth and donut family rather than choosing a window after reconstruction. Longer calendar-year and two-year outcomes test persistence, not the same local estimand.

For the timing analysis, marketing duration is the interval from posting to platform completion. A separate attention file is linked to the completed archive only through unique, outcome-blind exact matches. The linked sample contains 855 archive-observed sales for physical showings and duration and 755 for page views. All attention outcomes are secondary because they condition on observed completion; cumulative outcomes also mechanically load on exposure time.

4 Empirical Strategy: A Seasonal Triple Difference

4.1 Estimand and comparison groups

Index communities by c, cutoff experiments by $y \in \{2018, 2019\}$, and sides of January 1 by $s \in \{\text{before}, \text{after}\}$. Each side contains sixteen weeks. Let G_c indicate a formerly key-school community, R_y the real 2019 cutoff experiment, and P_s the post-January 1 side. For outcome Y_{cys}, the estimating equation is

$$\operatorname{asinh}(Y_{cys}) = \alpha_c + \lambda_{ys} + \rho(G_c \times R_y) + \phi(G_c \times P_s) + \tau_F(G_c \times R_y \times P_s) + \varepsilon_{cys}. \tag{2}$$

Community fixed effects absorb permanent level differences, and λ_{ys} absorbs every experiment-by-side shock common to exposed and control communities. The lower-order exposure interactions allow the groups to differ across experiments and in average seasonality. The coefficient is

$$\tau_F = [\Delta_{A-B}(G=1) - \Delta_{A-B}(G=0)]_{2019} - [\Delta_{A-B}(G=1) - \Delta_{A-B}(G=0)]_{2018}, \tag{3}$$

where Δ_{A-B} is the after-minus-before change. A negative τ_F means that the exposed-control seasonal difference moved downward more around the 2019 cutoff than around the identical 2018 calendar date; it does not by itself imply that exposed counts fell in absolute terms. I also estimate the count model by PPML with the same design, weights, and fixed effects. Its reported relative post ratio is $\exp(\widehat{\tau}_F^{\text{PPML}})$.

4.2 Identifying assumption and inference

The identifying restriction is parallel differential seasonality: absent the reform, the exposed-control difference in the sixteen-week winter-to-spring change in 2019 would equal the corresponding difference in 2018. This restriction is more credible than a raw before-after comparison because housing activity is highly seasonal (Ngai and Tenreyro, 2014). It is less demanding than assuming exposed and control communities have equal levels or equal unconditional trends. The successively broader archive benchmarks establish that completed exchange rose broadly in the policy-year window, but they do not supply this counterfactual or repair instability in the 2018 seasonal comparison.

Five threats remain. First, another January 2019 shock could differentially affect precisely the seven key schools. Second, the processed archive could undergo a treatment-correlated coverage change. Third, the historical school map could be wrong. Fourth, ordinary-school communities could receive spillovers from the same districtwide reform. Fifth, the platform completion date may misalign with administrative certification, including through the concurrent integrated-registration change described above. The design addresses, but cannot eliminate, these concerns through a prior-year calendar experiment, rolling false cutoffs, notice verification, both timing endpoints, local geographic controls, school-cluster inference, leave-one-community and leave-one-school estimates, and bandwidth/donut windows. For the fourth threat, Section 5.2 finds no detectable larger completed-flow increase among control communities nearest the exposed group, while leaving diffuse or delayed reallocation unresolved.

The control group is not untreated in the policy sense. Both groups entered multi-school zoning. The estimand contrasts replacement of a prior single-school mapping attached to a key school with replacement of an ordinary-school mapping. The DDD is a relative-exposure contrast that can combine exposed underperformance with any reform-induced gain or displaced activity in ordinary communities. It is not the district-average effect. Spillovers that move ordinary controls in the same direction attenuate the differential; spillovers of the opposite sign can amplify it.

Two timing predictions were fixed before reconstruction of missing completion dates: announcement-period acceleration and the sixteen-week cutoff comparison. The former is unsupported; the latter has the predicted sign under both completion-date endpoints and notice restrictions. The online appendix reports the complete timestamped protocol, including specifications that do not meet the original thresholds. These protocols were internal rather than externally preregistered.

4.3 Outcome-blind spatial validation

An outcome-blind spatial design replaces the original many-to-one frame with a global maximum-cardinality, minimum-total-distance assignment that never uses transaction outcomes. It pairs all nineteen exposed communities without replacement to distinct notice-verified controls within 2.384 kilometers and gives each of the seven exposed schools equal weight. This is a local-counterfactual validation, not a boundary discontinuity or a second source of policy variation: the data contain community points rather than historical catchment polygons, and nearby controls may receive displaced transactions. The online appendix reports the pair map, construction, calipers, road-network and geographic-surface checks, Lunar-New-Year exclusions, and pre-policy diagnostics (Butts, 2023).

5 Completed Exchange at the Title Reset

The headline result is a short-run divergence in completed flow. This section documents that gap and asks how much it depends on the seasonal benchmark. The broad announcement-to-cutoff acceleration prediction is unsupported at either completion-date endpoint. The surviving pattern develops over eight to sixteen weeks, consistent with search and administrative processing rather than a literal one-day discontinuity.

5.1 Formerly key communities do not join the 2019 spring rise

The policy-year pattern is visible beyond the selected comparison group. Across the same sixteen-week windows, completed transactions rise in Beijing outside Haidian, in Haidian communities outside the fifty-nine-community frame, and most strongly in matched ordinary-school communities; formerly key-school communities remain approximately flat. The 2019 after-to-before ratios are 0.96–1.08 in formerly key communities, 1.67–1.79 in ordinary communities, 1.20–1.33 in other Haidian, and 1.29–1.47 outside Haidian. These mutually exclusive groups are descriptive benchmarks from the same archive, not independent replications, assignment-verified controls, or untreated estimates. They document the observed rise against which the exposed segment's flat path is economically salient.

Figure 1 then shows the matched result before regression adjustment. In the preferred sample, 2019 design-weighted means move from 6.37 to 6.11 and from 6.00 to 6.47 in formerly key-school communities, compared with 2.91 to 4.87 and 2.79 to 4.99 in ordinary-school communities. The exposed counts therefore do not show a large observed contraction. In the identical-calendar 2018 comparison, both groups rise, with a larger increase in formerly key-school communities. The difference between these movements produces the relative regression result.

Table 1 reports the completed-flow DDD. Relative to the designated 2018 differential-seasonality benchmark, the asinh coefficients are −0.918 and −0.762, and separate PPML estimates imply ratios of 0.422 and 0.462. Under that one-year benchmark, these ratios represent exposed-relative gaps of 54–58 percent (Section 5.2 shows how the magnitude, though not the sign, depends on the benchmark year); they do not imply that exposed counts or total Haidian transactions fell by those percentages.

Exhaustively enumerated restricted-null school-wild tests for the asinh coefficients yield p-values of 0.012 and 0.035 across the two date conventions. CR2/Satterthwaite inference is more conservative at the upper endpoint, where the p-value is 0.062. I therefore emphasize agreement across completion-date conventions under the designated benchmark while retaining endpoint-specific uncertainty. The PPML ratios provide an interpretable scale; their conventional school-cluster tests are asymptotic and are not used for few-cluster significance claims. Literal notice-text controls, retained without rematching, imply similarly large relative gaps of 56.6 and 51.7 percent.

Table 1. Completed transaction flow around the January 2019 title-vintage cutoff

	Lower completion date		Upper completion date	
	Before	After	Before	After
Panel A. Raw completed transactions per community				
2018 identical-calendar comparison				
Formerly key-school communities	6.89	10.21	6.47	10.63
Ordinary-school comparison communities	3.16	3.44	2.92	3.68
2019 policy experiment				
Formerly key-school communities	6.37	6.11	6.00	6.47
Ordinary-school comparison communities	2.91	4.87	2.79	4.99
Observed 2019 change: formerly key-school (percent)	-4.1		7.9	
Observed 2019 change: ordinary-school (percent)	67.1		79.2	
Panel B. Relative, seasonally netted regression estimates				
Former key-school × After Jan. 1 × 2019	–0.918**		–0.762**	
CR2 standard error	(0.296)		(0.345)	
Restricted-null school-wild p-value	[0.012]		[0.035]	
CR2/Satterthwaite p-value	[0.017]		[0.062]	
PPML relative completed-flow ratio	0.422**		0.462**	
School-clustered PPML p-value	[0.043]		[0.037]	
Transactions (community–window cells)	1,173 (236)		1,173 (236)	
Panel C. Literal-notice assignment sensitivity				
Exact-notice controls: asinh DDD	–0.870**		–0.715*	
Exact-notice controls: school-wild p-value	[0.022]		[0.056]	
Exact-notice controls: PPML relative-flow ratio	0.434*		0.483*	

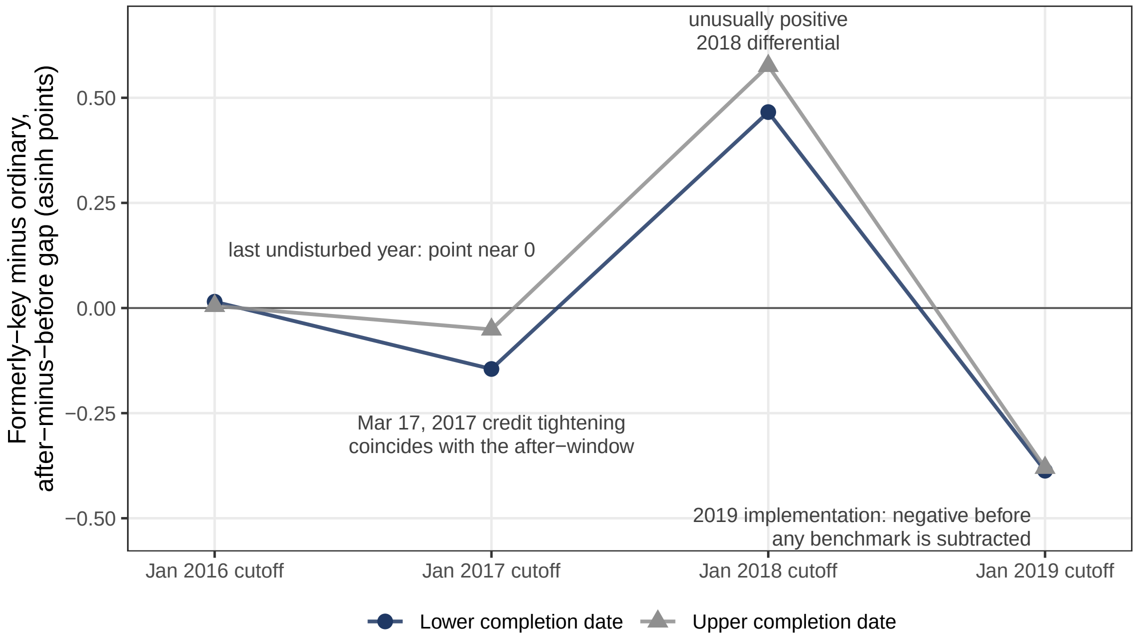
Notes: Each period contains sixteen weeks. Panel A reports inverse-reuse-weighted raw means for the 19 formerly key and 40 ordinary communities. Panel B reports the seasonally netted asinh DDD and a PPML ratio from the preferred specification with community and experiment-by-side fixed effects. Negative coefficients and ratios below one denote a relative shortfall, not a decline in formerly-key levels. Parentheses contain CR2 standard errors; brackets contain two-sided p-values. Stars use assigned-school tests: $^{***}p < .01$, $^{**}p < .05$, and $^{*}p < .10$. Lower and upper are the endpoints of the one-day completion-date interval. Panel C retains literal notice-text controls without rematching; full specification and inference details appear in the online appendix.

5.2 What the seasonal counterfactual identifies

The most consequential diagnostic is the seasonal benchmark. Figure 4 shows that 2018 is unusually positive: subtracting it from the negative 2019 gap produces the largest fixed-weight estimate, −0.85 to −0.96. The fully held-out 2018-versus-2017 contrast is sizeable and opposite-signed (0.61 and 0.63; wild $p = 0.068$ and 0.049). Benchmarks that predate or average over this oscillation yield smaller estimates of −0.24 to −0.64, while the raw 2019 gap is −0.39 before any benchmark is subtracted. These multiyear comparisons were added after observing the adverse placebo and are diagnostic rather than pre-specified. Every comparison preserves the negative sign; the credible magnitude is benchmark-sensitive.

Other diagnostics do not overturn the sign. Every community and assigned-school leave-out remains negative, all ten pre-specified bandwidth and donut specifications are negative, and the outcome-blind spatial match yields −0.72 at either endpoint. The spatial estimate weakens at a 1.5 kilometer caliper and under quadratic geographic surfaces. These checks limit particular composition concerns but do not repair the one-date or differential-seasonality assumptions; the online appendix reports the complete families without significance filtering. An exhaustive fixed-weight relabeling diagnostic points the same way. Holding the frozen frame, weights, and fixed effects unchanged, the actual seven-school labeling ranks 276th and 1,743rd from the most negative coefficient among all 116,280 labelings. Descriptive two-sided tail shares are 0.026 and 0.049 for the coefficient and 0.011 and 0.042 after CR1 studentization. Because key-school status was not assigned and the frame and weights were constructed around the actual exposure map, these shares are not randomization p-values or independent confirmation of the wild bootstrap. The online appendix reports the full distributions and absolute-value comparisons.

Figure 4. Formerly-key minus ordinary seasonal gaps by benchmark year



Notes: Each point is the formerly-key-minus-ordinary after-minus-before gap in completed transactions (asinh points) in a fixed-weight spatial frame chosen only from January 2015–August 2016 flows and coordinates. The 2018 point is unusually positive; 2016 is the last undisturbed year, with a point estimate near zero. The 2019 gap is negative before any benchmark is subtracted.

6 A Match-Threshold Model of Trade, Timing, and Selected Prices

The evidence to be organized is a large short-window relative-flow gap, slower selected completion, and small conditional-price point estimates. The model asks when a right attached to non-sale becomes a wedge to trade. Its organizing object is the bilateral match threshold: a title reset can raise that threshold, remove marginal matches, slow finite-horizon clearance, and change selection into observed prices. The model does not estimate the wedge. It shows which joint patterns one wedge can organize and how alternative attachment rules change the match set. The framework is deliberately partial equilibrium and uses standard housing-search and bargaining logic (Wheaton, 1990; Genesove and Han, 2012; Head et al., 2014; Han and Strange, 2015).

6.1 Sale raises the bilateral match threshold

Consider a homeowner deciding whether to retain a dwelling and a prospective buyer considering the same dwelling. Their reservation value and willingness to pay are

$$R_s = \bar{R}_s + a_s q_0, \tag{4}$$

$$V_b = \bar{V}_b + a_b \mu_1, \tag{5}$$

where q_0 is the value of the prior single-school mapping conditional on enrollment eligibility, $\mu_1 = \mathbb{E}[Q_1]$ is the expected value of the post-transfer assignment bundle, and $a_s, a_b \geq 0$ are household-specific school-use values. Moving costs, housing-match quality, and outside options are contained in $\bar{R}_s$ and $\bar{V}_b$. Define entitlement-free match surplus $X = \bar{V}_b - \bar{R}_s$ and the match threshold under title-contingent grandfathering

$$\Delta^T = a_s q_0 - a_b \mu_1. \tag{6}$$

The threshold has a useful decomposition:

$$\Delta^T = \underbrace{(a_s - a_b) q_0}_{\text{ordinary heterogeneity in use values}} + \underbrace{a_b (q_0 - \mu_1)}_{\text{buyer-side title-reset component}}. \tag{7}$$

The first term can arise even when an amenity is fully transferable. The second arises because the assignment bundled with the property changes when title transfers. It is this second component that distinguishes the policy from an ordinary capitalization problem. A bilateral match has nonnegative surplus if and only if

$$X \geq \Delta^T. \tag{8}$$

The superscript T denotes title-contingent grandfathering. The same physical house can therefore be mutually acceptable when the school right transfers and unacceptable when a sale resets eligibility. By contrast, an amenity change that shifts R_s and V_b by the same amount is capitalized into the transaction-price level while leaving match surplus, and hence equation (8), unchanged. That distinction separates the paper's transaction wedge from a standard capitalization exercise.

Conditional on predetermined property and market states, let X have cumulative distribution F, density f, and survivor function $\bar{F} = 1 - F$.

Proposition 1 (A higher threshold removes marginal matches). *The probability that a potential match has nonnegative surplus is*

$$p(\Delta) = \mathbb{P}(X \geq \Delta) = \bar{F}(\Delta). \tag{9}$$

At an interior point with $f(\Delta) > 0$, $p'(\Delta) = -f(\Delta) < 0$. An increase from Δ_0 to Δ_1 removes the set $\{X : \Delta_0 \leq X < \Delta_1\}$ from the positive-surplus region.

Proof. Differentiate $1 - F(\Delta)$ with respect to Δ. The discrete change is $F(\Delta_1) - F(\Delta_0)$. □

The model delivers a cross-sectional prediction only after an exposure ordering is supplied. If replacing a prior key-school mapping produces a larger expected loss than replacing an ordinary-school mapping, the post-transfer wedge rises more in the formerly key-school group. Completed activity in those communities can then fall behind ordinary-school communities. The empirical indicator supplies this ordering; it is not a household-specific estimate of Δ^T.

Remark 1 (Location composition during a common expansion)**.** *Index mutually exclusive location groups by $g \in \mathcal{G}$. Let $M_g > 0$ denote the mass of potential bilateral match opportunities in a fixed window. Expected accepted flow and its location share are*

$$C_g = M_g \bar{F}_g(\Delta_g), \qquad s_g = \frac{C_g}{\sum_{j \in \mathcal{G}} C_j}. \tag{10}$$

For the formerly-key group K, holding opportunity masses, surplus distributions, and other groups' thresholds fixed,

$$\frac{\partial s_K}{\partial \Delta_K} = -s_K(1 - s_K) h_{X,K}(\Delta_K) < 0. \tag{11}$$

Now consider a common expansion that scales every group's opportunity mass by $\rho > 1$. If the formerly-key threshold rises from Δ_{K0} to Δ_{K1} while other thresholds remain fixed, then

$$\frac{C_{K1}}{C_{K0}} = \rho \frac{\bar{F}_K(\Delta_{K1})}{\bar{F}_K(\Delta_{K0})} < \rho = \frac{C_{g1}}{C_{g0}}, \qquad g \neq K. \tag{12}$$

Accepted flow in formerly-key locations can therefore remain flat while flow elsewhere rises if the acceptance loss offsets the common expansion. The fall in s_K concerns the location composition of accepted matches. It neither requires nor establishes that an individual buyer switches from a formerly-key to an ordinary community. If opportunity masses, seller entry, or growth rates respond differentially across locations, equations (10) *and* (12) *remain a conditional benchmark rather than a structural decomposition. If comparison-group thresholds also change, the same relative conclusion requires the proportional decline in $\bar{F}_g(\Delta_g)$ to be larger in the formerly-key group; the exposure indicator alone does not establish that ordering structurally.*

6.2 Completed-flow responses are signed; selected-price responses are not

Suppose successful matches divide surplus through Nash bargaining and the seller receives share $\theta \in (0, 1)$. The transaction price is

$$P = R_s + \theta(V_b - R_s) = R_s + \theta(X - \Delta). \tag{13}$$

Prices are observed only when $X \geq \Delta$. Let $m(\Delta) = \mathbb{E}[X \mid X \geq \Delta]$, $e(\Delta) = m(\Delta) - \Delta$ denote mean excess surplus, and $h_X(\Delta) = f(\Delta)/\bar{F}(\Delta)$ denote the hazard of X.

Proposition 2 (The trade response is signed; the selected-price response is not)**.** *Holding R_s and θ fixed,*

$$\mathbb{E}[P \mid X \geq \Delta] = R_s + \theta e(\Delta), \tag{14}$$

$$\frac{\partial \mathbb{E}[P \mid X \geq \Delta]}{\partial \Delta} = \theta\{h_X(\Delta)e(\Delta) - 1\}. \tag{15}$$

Thus a larger wedge reduces the mass of acceptable matches but can lower, leave unchanged, or raise the mean price among completed trades.

Proof. Differentiating the conditional tail mean gives $m'(\Delta) = h_X(\Delta)[m(\Delta) - \Delta] = h_X(\Delta)e(\Delta)$. Because $e'(\Delta) = m'(\Delta) - 1$, substitution into equation (13) gives equation (15). □

For an exponential surplus distribution, mean excess surplus is constant and the selected mean price does not move locally even though acceptable matches disappear. Other distributions can generate either sign. A completed-sale hedonic regression therefore cannot recover the transaction wedge by itself, and controlling completed flow for a post-policy transaction price would condition on an equilibrium outcome. This derivative is an organizing comparative static, not a structural interpretation of the empirical triple difference.

6.3 A higher threshold slows finite-horizon clearance

Suppose potential buyers arrive at Poisson rate λ and draw independent match surpluses from F. Conditional on a property being continuously offered, the clearance rate is

$$h_C(\Delta) = \lambda \bar{F}(\Delta), \tag{16}$$

and the probability of completion by horizon T is

$$H(T, \Delta) = 1 - \exp\{-\lambda \bar{F}(\Delta)T\}. \tag{17}$$

Proposition 3 (A wedge delays trade without requiring permanent missing trade)**.** *For fixed λ and $T > 0$, a larger Δ lowers both the clearance rate and finite-horizon completion probability:*

$$\frac{\partial H(T, \Delta)}{\partial \Delta} = -\lambda T f(\Delta) \exp\{-\lambda \bar{F}(\Delta)T\} < 0. \tag{18}$$

Expected untruncated time to clearance is $1/h_C(\Delta)$ and rises with Δ. If two groups have strictly positive clearance rates, their cumulative-completion ratio converges to one as $T \to \infty$.

Proof. Differentiate equations (16) and (17). The limit follows because $H(T, \Delta) \to 1$ for every strictly positive $h_C(\Delta)$. □

This benchmark produces a sharp empirical implication: a large short-window divergence and a longer marketing clock can coexist with little cumulative deficit at a sufficiently long horizon. It does not assert that every property remains continuously listed or that all trades eventually occur. Withdrawals, seller entry, changing beliefs, and spatial reallocation can generate the same aggregate pattern, so the two-year result is a separately benchmarked qualitative consistency check, not an estimate or test of convergence in $H(T,\Delta)$.

To make that boundary explicit, let $L(\Delta)$ denote listing inflow and $W(\Delta)$ the withdrawal hazard. Active offered stock S evolves as

$$\dot{S} = L(\Delta) - [h_C(\Delta) + W(\Delta)]S, \qquad S^*(\Delta) = \frac{L(\Delta)}{h_C(\Delta) + W(\Delta)}. \tag{19}$$

The sign of the stock response is unrestricted: a wedge can reduce seller entry and clearance simultaneously. A stock snapshot therefore cannot separate entry, acceptance, and withdrawal, while a completed-transaction archive contains exits but no full-market risk-set denominator.

6.4 One threshold organizes quantity, timing, and selected prices

Table 2 makes the fact-to-model mapping explicit. The model is asked to organize the joint configuration, not to convert each point estimate into a structural parameter.

An exponential benchmark nevertheless gives the flow ratio an illustrative scale. Under its memoryless survivor function, $\bar{F}(\Delta) = \exp(-\Delta/\sigma)$, mean excess surplus among successful matches is σ and the selected mean price is invariant to the threshold when R_s and θ are fixed. This is a convenient sufficient benchmark, not a distribution selected by the imprecisely estimated price null. If opportunity masses are unchanged and the PPML DDD is read as an acceptance-rate ratio, ratios of 0.42–0.46 map into a differential threshold increase of 0.77–0.87 mean excess-surplus units. Without the exponential assumption, the same maintained acceptance-rate mapping says that 54–58 percent fewer matches clear relative to the seasonal counterfactual. These are assumption-indexed normalizations of the relative flow estimate, not estimates of a monetary wedge or welfare loss.

Seller reservation values, aggregate demand, seller entry, withdrawal, and congestion can produce similar flow–price–duration patterns (Stein, 1995; Genesove and Mayer, 2001; Anenberg and Ringo, 2024, 2026). The model therefore disciplines an economic interpretation rather than identifying a unique primitive. What is setting-specific is the legal source of the transfer-contingent asymmetry.

Table 2. Empirical facts and their model counterparts

Empirical fact	Model object	Discipline and remaining margin
Completed-flow ratio 0.42–0.46	Acceptance and clearance, $\bar{F}(\Delta)$ and $h_C(\Delta)$	Fewer or delayed marginal matches; buyer arrival and seller entry remain combined
Flat formerly-key flow during a broad expansion	Accepted-location share s_K	A higher relative threshold can shift accepted flow; this is not buyer substitution
Conditional price and discount coefficients near zero	Selected mean in equation (15)	Price is not sufficient for the wedge; the entitlement value remains unidentified
Duration points rise; posting ratio 0.96; stock ratios 0.83–0.92	Clearance clock and stock law	Directionally consistent with slower execution across distinct risk sets
Cumulative two-year ratio 0.945	Finite-horizon completion	Consistent with timing or reallocation; not property-level convergence

Notes: The table maps empirical objects to model counterparts. It does not treat the estimates as moments from a common risk set or use them to recover structural parameters.

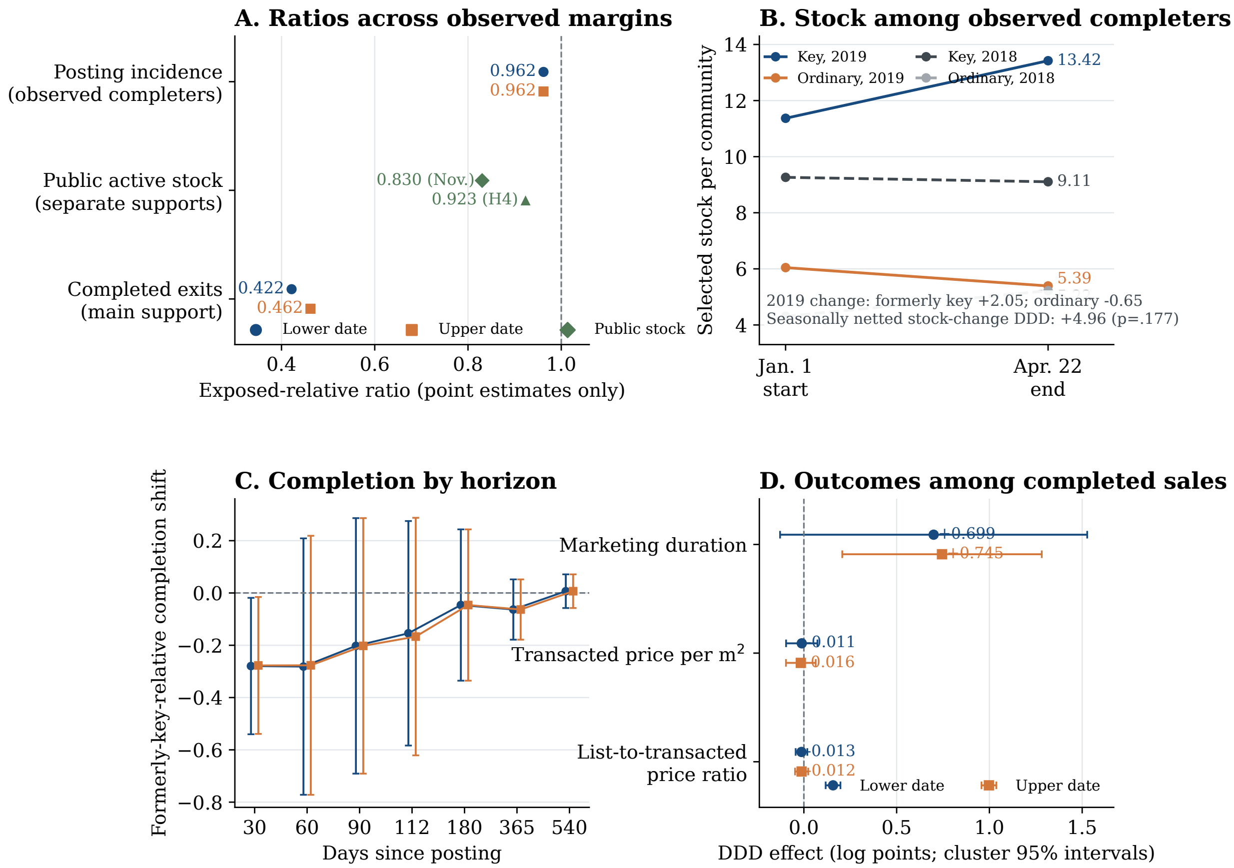
7 Inside the Exchange Funnel

The model yields four observable implications of a higher relative match threshold: fewer completed exits, slower clearance, more active stock if entry does not contract equally, and an unsigned price response among selected trades. I examine these implications across three observation universes. Posting and completion clocks provide exact accounting within listings observed to complete by the archive end; independent public crawls provide point observations of active stock; and the completed-sale archive supplies duration and price outcomes. Their denominators differ, so the evidence disciplines a joint configuration rather than an additive decomposition of the offered market. Figure 5 summarizes the sequence; the subsections then define each risk set and estimand.

7.1 Posting, stock, and exit among observed completers

The archive does not observe the full offered universe, but posting and completion dates permit exact stock-flow accounting within listings observed to transact by the archive end. This selected pipeline contains 1,364 posting events and 1,173 completed exits across 236 cells, and its accounting identity has zero error under both completion-date conventions. Posting incidence has an exposed-relative PPML ratio of 0.962 under both dates, whereas exits reproduce the headline ratios of 0.422 and 0.462. In the 2019 after window, selected stock contracts by 0.65–0.78 observed completers per ordinary community but expands by 1.68–2.05 per formerly key community. The seasonally netted stock-change point estimate is +4.84 to +4.96 successful listings per community (p = 0.225 and 0.177). This is exact

Figure 5. Posting, selected stock, completion timing, and completed-sale outcomes



Notes: Panel A reports point ratios for posting incidence among archive-observed completers, public active stock, and completed exits; the horizontal positions are not confidence intervals. Panel B gives exact selected-pipeline stock accounting. Panel C fixes posting cohorts but conditions on completion by the archive end. Panel D conditions on completed sales and shows conventional assigned-school cluster intervals. Lower and upper markers denote the two feasible completion-date endpoints. Panels use different risk sets and do not form an additive decomposition.

accounting within successful spells, not observed full-market inventory. The posting ratio is a point estimate, not an equivalence result; because the pipeline conditions on completion by the archive end, it does not measure full-market listing entry or a sale hazard.

A fixed-posting-cohort diagnostic describes timing inside this selected set. Among 709 listings observed to complete by the archive end and posted from January 1 through April 22 in 2018 or 2019, the formerly-key-relative change in completion probability is −0.280 and −0.277 at thirty days. CR2 and wild p-values are approximately 0.04, but no horizon survives a Holm adjustment across seven horizons. By day 112 the point difference narrows to −0.15 to −0.17 and is imprecise. Both cohorts have at least 540 days of follow-up; the 2019 exposed cohort represents only five schools, and convergence at long horizons is mechanical within a sample selected to complete. The online appendix reports the entire curve, every horizon, both unadjusted inference procedures, and the family-wise conclusion.

7.2 Public active stock and an assumption-indexed point calibration

Public active-stock snapshots offer an independent comparison. On identical community supports, their exposed-relative point ratios lie closer to one than the corresponding completed-flow ratios, but the waves do not form a listing-transition panel. On the November-source support, the stock point is 0.830 while completed-flow ratios are 0.420 and 0.464. On the H4 support, the stock point is 0.923 while flow ratios are 0.368 and 0.427. The point comparisons therefore differ sharply from completed flow without identifying a common stock–flow estimand.

The comparison is an assumption-indexed point calibration rather than a causal decomposition or confidence set. The online appendix writes completed flow as period exposure times execution and defines a multiplier κ for cross-crawler coverage, the unobserved pre-period stock ratio, and point-stock-to-period-exposure alignment. At $\kappa = 1$, the points imply execution-rate ratios of approximately 0.40–0.56; eliminating the execution difference at the point estimates requires κ between 1.79 and 2.51. The exercise makes the required stock adjustment transparent but does not separate buyer arrival, match acceptance, withdrawal, listing entry, or bargaining.

7.3 Duration and selected prices

Under the designated benchmark, marketing duration among completed sales rises 0.70 and 0.75 log points at the two date endpoints (wild $p = 0.238$ and 0.043; Holm-corrected 0.714 and 0.130 across the three conditional outcomes). Transacted prices per square meter move −0.011 and −0.016 log points, and list-to-transacted-price ratios move −0.013 and −0.012; neither price family is statistically distinguishable from zero. The online appendix reports the complete pre-specified outcome table and a tail diagnostic that is directionally consistent with more unusually slow completed trades but imprecise at both endpoints.

Among 855 eventual sales with exact completion dates and unique outcome-blind links to the attention archive, marketing duration rises by 0.857 log-one-plus points (school-wild $p = 0.019$; CR2/Satterthwaite $p = 0.035$). Cumulative physical showings also rise, but daily follower, page-view, and physical-showing outcomes are imprecise. The scale-corrected physical-showing-rate ratio is 0.728 ($p = 0.497$ against one), too imprecise to establish preserved buyer arrival.

The configuration is consistent with the model's selection-and-delay pattern, but it does not identify a unique mechanism. Equation (15) shows why removing marginal matches need not move the mean price among surviving trades. Because duration conditions on observed completion and the flow window truncates later exits, the data do not support a quantitative mapping between duration and completed-flow ratios. A group-specific demand, amenity, expectations, or platform-coverage shock could also reduce relative completed flow. The small conditional price estimates do not rule out those alternatives, especially with confidence intervals of roughly ±8 percent. They establish a narrower point: conditional transaction prices need not move in proportion to completed flow.

7.4 Which units carry the gap

Unit type is the archive's only proxy for the kinds of households valuing the school right; it does not observe buyer composition. Two readings were recorded before this split was estimated. If the right matters mainly to families intending to use the school, the relative gap may be larger for family-sized units. If school-motivated purchases instead concentrate in small units bought mainly for the right—the placeholder-home reading familiar from Beijing's school-district market—the gap may be larger for small units. Table 3 reports splits at 60 square meters, a round threshold near the 2019 window's first quartile, and at one versus two or more bedrooms.

Under the designated 2018 benchmark, weighted-asinh estimates are more negative above 60 square meters: −0.90 to −0.96, compared with −0.15 to −0.24 below the threshold, with unadjusted interaction p-values of 0.050–0.052 (0.20–0.21 after adjustment across four benchmark comparisons). This contrast is not a clean mechanism test. At the upper endpoint, the raw 2019 seasonal DD is −0.339 in both size classes; the interaction is generated by their different 2018 benchmarks. Comparisons with 2016 and 2017 attenuate the large-unit differential. The bedroom split is also estimator-sensitive: weighted-asinh estimates are more negative for two-plus-bedroom units, whereas PPML ratios are slightly lower for one-bedroom units. I therefore treat the splits as suggestive benchmark-specific heterogeneity, not evidence that identifies family buyers or rules out the placeholder-home and broader demand-shock alternatives.

Table 3. Benchmark-specific completed-flow heterogeneity by unit type

	Asinh DDD (s.e.)		Wild p [Holm]		PPML ratio	Trans.
	Lower	Upper	Lower	Upper	Lower / Upper	
Area ≤ 60 m^2	−0.237 (0.330)	−0.152 (0.373)	0.467 [0.467]	0.676 [0.676]	0.602 / 0.611	262
Area > 60 m^2	−0.963 (0.318)	−0.904 (0.333)	0.004 [0.007]	0.002 [0.003]	0.398 / 0.447	911
small minus large	+0.726 (0.343)	+0.752 (0.384)	0.052	0.050		
One bedroom	−0.180 (0.261)	−0.189 (0.281)	0.449 [0.449]	0.470 [0.470]	0.409 / 0.455	206
Two-plus bedrooms	−0.850 (0.297)	−0.786 (0.313)	0.006 [0.013]	0.003 [0.006]	0.452 / 0.502	960
one minus two-plus	+0.669 (0.299)	+0.596 (0.335)	0.033	0.048		

Notes: Unit-type split of the designated 2019-versus-2018 design (Section 7.4). Cells are community × experiment × side × unit class; each class-specific row re-estimates the weighted-asinh DDD with community and experiment-side fixed effects, inverse-reuse weights, and assigned-school clusters. Difference rows come from pooled models with class-specific fixed effects. Wild-school p-values use 9,999 restricted-null studentized draws; brackets apply Holm correction across the two class-specific tests within each split, not across the interaction tests. The 60 m^2 cutoff was fixed as a round threshold near the 2019 window's first quartile and classifies 262 of 1,173 observations as small. The bedroom split omits seven observations with missing room counts. PPML ordering supports the larger-unit reading for floor area but not for bedrooms; small-unit cells hold roughly one transaction each.

8 When and Where the Gap Attenuates

The short-window result is not a permanent market-freeze estimate. The broad announcement-to-cutoff acceleration test is unsupported, and two separately defined diagnostics ask whether the gap remains visible over longer time and within local pairs.

8.1 A separate cumulative comparison

A separately locked persistence exercise yields an exposed-relative PPML ratio of 0.84 for calendar 2019 and 1.01–1.02 for calendar 2020. For cumulative 2019–2020 flow relative to 2016–2017, the ratio is 0.945 at both date endpoints, with asinh wild $p = 0.815$ and 0.835. No cumulative difference is detectable in this separately benchmarked comparison. This is consistent with delay or reallocation, but it is not a within-property convergence estimate; a similarly sized 2017 placebo also limits its use as causal validation. The online appendix reports the complete horizon table.

8.2 Local rotation is suggestive, not identified substitution

Displacement into ordinary-school controls is the sharpest spillover concern. Pre-specified geographic diagnostics find no larger increase among nearby controls, and far-control and shared-bizcircle exclusions leave PPML ratios of 0.41–0.47. Within the nineteen outcome-blind spatial pairs, combined flow shows no detectable differential while the formerly-key share falls by an imprecisely estimated 10.4–11.5 percentage points. The composition shift is consistent with local rotation but does not establish substitution: the pairs are not closed markets, and diffuse or delayed reallocation remains one explanation for the cumulative two-year result. The online appendix reports the complete proximity tests and pair decomposition.

9 Transition Design

The policy tension is not grandfathering itself but protection conditioned on non-sale. Under title-contingent grandfathering, retaining the asset preserves the old rule and transfer extinguishes it. The model separates that reliance objective from the attachment rule: protection can instead follow the property, the household, or a common transition date.

9.1 Attachment rules and the exchange margin

Under title-contingent grandfathering, the wedge is Δ^T in equation (6). If the old assignment follows the property after sale, if the incumbent household can carry it after moving, or if old and new owners face a uniform post-transition rule, respectively, the corresponding wedges are

$$\Delta^P = a_s q_0 - a_b q_0, \tag{20}$$

$$\Delta^H = -a_b \mu_1, \tag{21}$$

$$\Delta^U = (a_s - a_b)\mu_1. \tag{22}$$

Property portability raises what the buyer obtains from the dwelling. Household portability removes the incumbent's lost entitlement from the reservation value of selling. A uniform transition removes the title-vintage difference, but changes the incumbent's protected bundle even without a sale.

Proposition 4 (Portability weakly expands the match set)**.** *Suppose $q_0 \geq \mu_1$ and $a_s, a_b \geq 0$. Holding school capacity, assignment probabilities, and outside housing options fixed,*

$$\Delta^T \geq \Delta^P, \qquad \Delta^T \geq \Delta^H, \qquad \Delta^T \geq \Delta^U. \tag{23}$$

Hence each portable or uniform rule weakly raises the bilateral match probability relative to title-contingent grandfathering. The ordering among P, H, and U is generally ambiguous.

Proof. The three differences are $\Delta^T - \Delta^P = a_b(q_0 - \mu_1)$, $\Delta^T - \Delta^H = a_s q_0$, and $\Delta^T - \Delta^U = a_s(q_0 - \mu_1)$, all weakly nonnegative. Apply Proposition 1. □

Proposition 4 isolates a narrow transition-design margin. It is a partial-equilibrium match-rate ordering, not a welfare ranking. Its design implication is narrower: honoring reliance does not require making sale the forfeiture event. Table 4 shows exactly which side of the match each rule changes.

Table 4. What sale changes under alternative attachment rules

Rule	What happens at sale	Exchange-margin result	Main unmeasured trade-off
Title-contingent	Incumbent loses q_0; buyer receives μ_1	Benchmark threshold $\Delta^T = a_s q_0 - a_b \mu_1$	Sorting into legacy titles; capacity attached to old addresses
Property portability	Assignment q_0 follows the dwelling	Lowers threshold by $a_b(q_0 - \mu_1)$	Perpetuates address-based advantage and capitalization
Household portability	Incumbent carries q_0; buyer receives μ_1	Lowers threshold by $a_s q_0$	Travel, cross-neighborhood use, and destination capacity
Uniform transition	Incumbent and buyer both face μ_1	Lowers threshold by $a_s(q_0 - \mu_1)$	Weakens reliance protection; changes congestion and incidence

Notes: Threshold differences follow from Proposition 4 under $q_0 \geq \mu_1$ and $a_s, a_b \geq 0$. They order bilateral match probabilities in partial equilibrium, not welfare. A welfare comparison additionally requires preferences, assignment probabilities, capacity, realized school outcomes, and equilibrium sorting.

Which portable rule is appropriate depends on what the transition is meant to protect. Property portability preserves the address-based entitlement and lets a buyer obtain it through the house price, but can perpetuate the spatial scarcity rent. Household portability protects the incumbent family while allowing the dwelling to enter the new regime, but can create travel and destination-capacity costs. A common date-based transition restores symmetry when equal treatment is valued more than legacy protection. A time-limited household entitlement is one implementation worth evaluating, not a rule selected by the model or the data.

9.2 Scale, mechanism, and scope

An external transfer-tax yardstick. A sale-contingent entitlement reset enters bilateral surplus like a heterogeneous transaction cost, although it is not a statutory tax, raises no public revenue, and varies with household demand for schools. Transfer-tax studies show that modest sale-contingent wedges can move transaction volume and timing sharply (Dachis et al., 2012; Best and Kleven, 2018). These estimates provide an external mechanism benchmark, not a conversion of the Haidian coefficient into an implicit tax rate. California's household-portability provision for Proposition 13, which reduced measured lock-in among newly eligible movers, illustrates a different attachment rule in practice (Ferreira, 2010).

What the evidence identifies. The data identify a short-run formerly-key–ordinary difference in completed flow, not the primitive generating it. The title clause supplies a specific transfer-contingent mechanism, and the selected price and duration pattern is consistent with that mechanism, but buyer arrival, seller entry, withdrawal, platform coverage, bargaining, and spatial reallocation remain observationally combined. The result also does not imply price rigidity: selection can move the mean price among completed trades in either direction. Welfare and a ranking of transition rules require assignment, preference, capacity, and migration data not observed here. Structural work integrating residential location and school choice illustrates the additional equilibrium objects such exercises require (Bayer et al., 2007; Cui et al., 2026).

External validity rests on the mechanism rather than the local average effect: whenever ownership bundles a scarce public benefit and transfer extinguishes that benefit, sale can destroy bilateral surplus. The magnitude estimated here remains local to nineteen communities attached to seven formerly key schools in one Beijing district; it is not the average effect of multi-school zoning on Haidian or Beijing.

10 Conclusion

Beijing's title-vintage rule made sale the event that extinguished a legacy school-assignment mapping. Around implementation, formerly key-school communities remained near six completions per community while ordinary-school communities rose from roughly three to five. Relative to the identical 2018 seasonal contrast, the completed-flow ratio is 0.42–0.46; earlier and averaged benchmarks preserve the sign but imply smaller gaps. Across distinct risk sets, posting among archive-observed completers moves little at the point estimate, selected-stock and early-completion estimates shift toward slower execution, active-stock points lie closer to one than completed flow, and selected transaction prices move little. No separately benchmarked two-year difference is detectable. The evidence therefore supports a short-run relative exchange gap, not a permanent market freeze or an identified full-market hazard.

A match-threshold model explains how a nonportable right removes or delays marginal trades without requiring a proportional change in the mean price of trades that complete. It also separates reliance protection from the rule that conditions protection on non-sale. Under the model's partial-equilibrium conditions, attaching protection to the household or property—or ending it on a common date—expands the feasible match set relative to a standing-title rule. This is not a welfare ranking, but it clarifies the policy choice: protecting reliance need not make exchange the event that destroys protection.

Two next tests would sharpen the mechanism. Xicheng's analogous rule for families obtaining title after July 31, 2020 is the natural external replication once post-2020 transactions can be observed (Xicheng District Education Commission, 2020; Shao et al., 2023). Linking complete offered-market histories—entry, price revisions, withdrawal, relisting, and sale—to title registration and realized school assignment would then separate delayed execution from missing trade and permit a quantitative evaluation of alternative transition rules.

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Online Appendix

Grandfathering at Sale: Nonportable School Rights and Housing Exchange

Tai Lo Yeung

September 2, 2026

A Model Details

Let the incumbent's reservation value and a prospective buyer's willingness to pay be

$$R_s = \bar{R}_s + a_s q_0, \tag{A1}$$

$$V_b = \bar{V}_b + a_b \mu_1, \tag{A2}$$

where q_0 is the deterministic entitlement preserved by retaining the old title, μ_1 is the expected value of the post-transfer assignment bundle, and $a_s, a_b \geq 0$ are household-specific school-use values. Define entitlement-free match surplus $X = \bar{V}_b - \bar{R}_s$ and the transaction-contingent wedge

$$\Delta = a_s q_0 - a_b \mu_1. \tag{A3}$$

Trade occurs if and only if $X \geq \Delta$.

Proposition A1 (A symmetric common shift leaves the model's trade set unchanged)**.** *If a policy changes buyer value and seller reservation value by the same amount, it changes the bargaining price but leaves the trade set unchanged.*

Proof. Adding the same scalar d to R_s and V_b leaves $V_b - R_s$ and hence the positive-surplus indicator unchanged. □

Holding the valuation loadings fixed and treating Δ as a scalar threshold, suppose X has cumulative distribution F and density f. The probability that a potential match trades is $p(\Delta) = 1 - F(\Delta)$.

Proposition A2 (A larger transaction-contingent wedge removes marginal matches)**.** *At any interior point with $f(\Delta) > 0$,*

$$\frac{\partial p(\Delta)}{\partial \Delta} = -f(\Delta) < 0. \tag{A4}$$

An increase from Δ_0 to Δ_1 removes the set of matches $\{X : \Delta_0 \leq X < \Delta_1\}$.

To characterize selection into observed prices, let the seller receive Nash weight θ and let $m(\Delta) = \mathbb{E}[X \mid X \geq \Delta]$. Conditional mean transaction price is

$$\mathbb{E}[P \mid X \geq \Delta] = R_s + \theta\{m(\Delta) - \Delta\}. \tag{A5}$$

Writing $e(\Delta) = m(\Delta) - \Delta$ and $h(\Delta) = f(\Delta)/[1 - F(\Delta)]$, and holding R_s and θ fixed, gives

$$\frac{\partial \mathbb{E}[P \mid X \geq \Delta]}{\partial \Delta} = \theta\{h(\Delta)e(\Delta) - 1\}. \tag{A6}$$

The selected-price response is unsigned even though match probability falls. If potential buyers arrive at rate λ, a listing that remains offered clears at rate $\lambda[1-F(\Delta)]$. A larger wedge lowers clearance and lengthens the expected marketing spell for fixed λ, but duration among listings observed to complete remains selected and active stock also depends on listing entry and withdrawal.

B Measurement and Assignment Audit

The canonical transaction input is a processed proprietary listing-platform export containing 586,312 Beijing resale records posted from 2010 through January 2021. The policy analysis uses 2015–2020. All retained rows have positive transaction price, posting date, and marketing duration, so the archive contains completed transactions rather than an offered-listing denominator. The provider export does not document time-varying platform market share or off-platform transactions. The distributable package begins with the canonical analysis file and preserves every downstream policy transformation; reproducing the upstream provider-file cleaning step requires redistribution permission and a verified raw-to-canonical crosswalk.

A separate provider auxiliary listing-and-attention export does contain rows without published transaction prices. Within the fifty-nine-community frame, however, only nineteen such rows have 2018 posting years and sixty-one have 2019 posting years; their observed status dates begin on April 23, 2019 and are asynchronous across records. These rows establish that the raw bundle is not limited to completed transactions, but they do not recover the offered stock at the January 1 implementation date or form a listing-level policy risk set. The paper therefore does not reinterpret their missing transaction prices as observed failures to sell.

When the exact completion date is missing because the source date is month-only, the paper retains the fixed interval from posting date plus reported duration minus one day to posting date plus duration. In the 414,979-record validation sample that observes both clocks, all but one reconstructed date lies within one day of the reported date. In the broader matched frame, 27 of 2,515 reconstructed dates (1.1 percent, or 0.36 percent of all 7,410 transactions) change cutoff-cell assignment across endpoints.

Table A1: Transaction archive and completion-date measurement

Panel A. Archive-wide validation

Record universe	Count
Beijing completed-transaction archive	586,312
Transactions observing exact and reconstructed clocks	414,979
Reconstructed dates changing cutoff windows (broader frame)	27 of 2,515 (1.1%)

Panel B. Retained-sample dates

	Preferred assignment frame	Exact-notice sensitivity
Completed transactions in four estimation windows	1,173	1,029
Resolved archive support in retained communities	7,340	6,519
Archive records with reported exact date	4,863	4,303
Archive records with reconstructed date	2,477	2,216

Notes: The archive consists entirely of completed transactions. Reported exact dates are retained. Missing exact dates use the two fixed endpoints implied by posting date and marketing duration. The 414,979-record validation observes both clocks; all but one reconstructed date is within one day of the reported completion date. The broader-frame cutoff-window statistic has denominator 2,515 reconstructed dates (0.36 percent of all 7,410 records change windows).

Figure A1 reports an unsmoothed calendar-month diagnostic under both endpoints of the fixed completion-date interval. It shows neither a broad announcement-to-cutoff acceleration nor a literal one-month break at January 1; the exposed–ordinary divergence in sixteen-week completed flow develops over the subsequent spring. The plot is deliberately not interpreted as an event study. Calendar-month assignment is much more sensitive to the one-day convention than the main sixteen-week cells: 232 reconstructed transactions move to an adjacent month between April 2018 and May 2019, whereas only 27 reconstructed transactions change membership in the main cutoff cells.

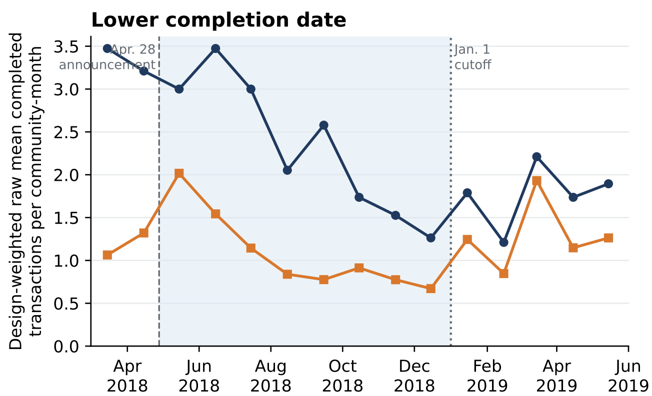

Figure A1: Raw monthly completed transactions under the two completion-date conventions

Notes: Each point is the inverse-reuse-weighted raw mean number of completed transactions per community-month in the preferred 59-community notice-verified frame. Zero-flow community-months are retained; no smoothing or significance filtering is applied. The two panels assign interval-dated transactions to the lower and upper endpoints of the fixed one-day completion interval. Raw group levels are not causal effects: both groups entered multi-school zoning, formerly key-school communities have higher baseline flow, and ordinary communities may receive reallocated transactions. April 2018 contains only three post-announcement days, and May 2019 lies beyond the formal sixteen-week post-cutoff window. Dates are platform-recorded completions rather than administrative title registrations.

The assignment frame uses visibly dated 2017 school-admission notices for every exposed community. The preferred comparison sample retains communities named literally in those notices or linked through an exact and unambiguous property or address alias. The exact-notice sensitivity retains only literal text matches. These restrictions preserve all nineteen exposed communities and do not rematch the design.

After freezing the design, I conducted an outcome-blind archival reconstruction of 2018 notice images to test whether the treated assignment was a transitory 2017 classification. Direct image review verifies the same assigned school in both years for all nineteen exposed communities and all seven exposed schools; Table A2 reports the school-level support. The 2018 images are preserved with page URLs and SHA-256 hashes, but are hosted by a public secondary archive rather than an authenticated district database. The main exposure therefore remains the frozen 2017 assignment, and the two-year check is used only as provenance and stability validation. Control-notice coverage is incomplete in 2018, so no control is added, deleted, or relabeled using that archive.

Table A2: Two-year historical-assignment validation for exposed communities

Historically assigned school	Exposed communities	2017 notice	Same assignment in 2018
Qiyi Primary School	3	Verified	3 of 3
Renda Affiliated Experimental Primary	1	Verified	1 of 1
Shangdi Experimental Primary School	3	Verified	3 of 3
CNU Affiliated Yuxin School	2	Verified	2 of 2
Yangfangdian Central Primary School	1	Verified	1 of 1
Zhongguancun No. 3 Primary School	1	Verified	1 of 1
Renda Affiliated Primary School	8	Verified	8 of 8
Total	19	19 of 19	19 of 19

Notes: The design is fixed from the 2017 notices. The 2018 column is a subsequent, outcome-blind stability audit based on direct review of dated notice images and does not rematch or select the estimation sample. Source pages, retained images, direct-read excerpts, and hashes are recorded in the replication archive. The images are hosted by a public secondary archive; the table therefore validates archival consistency, not district-database authentication.

Tables A3 and A4 provide stable row-level audit IDs and notice-image hash prefixes. The corresponding UTF-8 files in the replication folder preserve the original community and school names, matching phrase, public source URL, local image name, rationale, and full SHA-256 hash. Excluded candidates remain in the control audit rather than disappearing from the provenance trail.

Table A3: Exposed-community notice manifest

Audit ID	School ID	Longitude	Latitude	Notice page	SHA-256 prefix
HT01	3	116.290	39.967	11330	e3ce3d7c28
HT02	3	116.288	39.969	11330	e3ce3d7c28
HT03	3	116.289	39.965	11330	e3ce3d7c28
HT04	3	116.295	39.969	11330	e3ce3d7c28
HT05	3	116.292	39.972	11330	e3ce3d7c28
HT06	3	116.285	39.969	11330	e3ce3d7c28
HT07	3	116.292	39.969	11330	e3ce3d7c28
HT08	3	116.285	39.967	11330	e3ce3d7c28
HT09	4	116.298	39.977	11268	dadbcaffeb
HT10	30	116.358	40.071	12127	1927f52005
HT11	30	116.352	40.062	12127	1927f52005
HT12	46	116.338	39.914	11303	85eb12ddc1
HT13	59	116.320	40.039	11342	abf24ecb4b
HT14	59	116.323	40.041	11342	abf24ecb4b
HT15	59	116.318	40.039	11342	abf24ecb4b
HT16	61	116.305	39.895	11233	7e9d768cef
HT17	61	116.315	39.901	11233	7e9d768cef
HT18	61	116.313	39.902	11233	7e9d768cef
HT19	67	116.340	39.989	12175	d27f2ba414

Notes: Stable audit IDs map to the UTF-8 community and school names, complete notice URLs, dates, image filenames, and full SHA-256 hashes in `Replication/frozen_matched_community_frame_assignment_audited.csv`. All nineteen records are visibly dated 2017 notice-image links. School IDs are stable internal identifiers, not rankings.

Table A4: Ordinary-community assignment audit manifest

Audit ID	Evidence	Preferred	Literal	SHA-256 prefix
HC01	E	Yes	Yes	a3e6ae419b
HC02	U	No	No	5acd502cb8
HC03	E	Yes	Yes	5acd502cb8
HC04	E	Yes	Yes	5acd502cb8
HC05	A	Yes	No	5acd502cb8
HC06	E	Yes	Yes	5acd502cb8
HC07	E	Yes	Yes	5acd502cb8
HC08	E	Yes	Yes	5acd502cb8
HC09	E	Yes	Yes	5acd502cb8
HC10	E	Yes	Yes	b473d367d5
HC11	E	Yes	Yes	b473d367d5
HC12	E	Yes	Yes	b473d367d5
HC13	E	Yes	Yes	de7bd69caa
HC14	E	Yes	Yes	649ceaa638
HC15	E	Yes	Yes	649ceaa638
HC16	E	Yes	Yes	6cde492527
HC17	E	Yes	Yes	6cde492527
HC18	E	Yes	Yes	6cde492527
HC19	E	Yes	Yes	6cde492527
HC20	E	Yes	Yes	3016590a80
HC21	E	Yes	Yes	3016590a80
HC22	E	Yes	Yes	3016590a80
HC23	A	Yes	No	3016590a80
HC24	E	Yes	Yes	54886b022f
HC25	E	Yes	Yes	54886b022f
HC26	E	Yes	Yes	0376bdbe12
HC27	S	No	No	d7913db869
HC28	E	Yes	Yes	a8edee6900
HC29	E	Yes	Yes	a8edee6900
HC30	A	Yes	No	a8edee6900
HC31	E	Yes	Yes	754729ffe5
HC32	A	Yes	No	754729ffe5
HC33	A	Yes	No	754729ffe5
HC34	E	Yes	Yes	754729ffe5
HC35	A	Yes	No	754729ffe5
HC36	A	Yes	No	754729ffe5
HC37	E	Yes	Yes	d976c37825
HC38	A	Yes	No	3c7e243264
HC39	E	Yes	Yes	3c7e243264
HC40	E	Yes	Yes	3c7e243264
HC41	A	Yes	No	3c7e243264
HC42	E	Yes	Yes	446569c893

Notes: E = literal notice text; A = exact address or unambiguous alias; S = secondary source only; U = unresolved. The preferred frame retains 31 E and 9 A records; the literal-notice sensitivity retains only E. The two excluded candidates remain visible. The full UTF-8 crosswalk—including community and school names, matching phrases, source URLs, rationales, local image names, and complete hashes—is `Replication/control_assignment_audit_2017.csv`.

The external formerly-key classification also has independent pre-announcement market content. A school component estimated from 2015–2016 transaction prices improves 2017 out-of-sample prediction, and the seven formerly-key schools have a higher component than the fourteen ordinary schools. Table A5 reports this treatment-proxy validation. A separately frozen continuous-dose policy specification fails its promotion gates and is not used to rescue the binary result.

Table A5: Pre-announcement market-value validation of the binary exposure proxy

Validation quantity	Result
Historical schools with 2015–2016 support	21 / 21
Schools with at least five support transactions	18 / 21
2017 transaction-level MSE reduction from school component	4.67%
2017 school-mean validation slope (N = 20 schools)	0.934
One-sided validation-slope p-value	0.0000062
Formerly-key minus ordinary school-value dose	0.851 SD
One-sided enumerated classification-label sensitivity p-value	0.0462
Continuous policy-dose promotion gate	Fails

Notes: The school component is estimated only from 2015–2016 prices and validated on 2017 transactions before the policy outcome is opened. All twenty-one schools have training support; the validation-slope regression uses the twenty schools with 2017 transactions. The binary key-school classification is external to this exercise. The one-sided label sensitivity enumerates all 116,280 assignments of seven labels among twenty-one schools and is a treatment-proxy diagnostic, not randomization inference for the policy effect. The separately frozen continuous-dose policy interaction fails significance and placebo-dominance gates and is not used as an outcome specification.

Table A6 records the observation counts, communities, clusters, and identifying variation for the headline and supporting samples.

Table A6: Estimation samples and identifying support

	Preferred assignment frame	Exact-notice sensitivity
Completed transactions in four estimation windows	1,173	1,029
Exposed communities	19	19
Comparison communities	40	31
Matched sets	19	19
Historically assigned-school clusters	21	21
Exposed-school clusters	7	7
Community–experiment–side cells	236	200

Notes: The preferred frame retains comparison communities named in contemporaneous 2017 notices or linked by an exact, unambiguous address alias. The exact-notice sensitivity retains only literal text matches. Both samples preserve all nineteen exposed communities and seven exposed schools. Each community contributes four observations: before and after January 1 in the 2018 seasonal comparison and the 2019 policy experiment.

C Additional Identification and Robustness Evidence

Before reconstructing missing completion dates, I recorded two designs in timestamped internal protocols: an announcement-period acceleration and a sixteen-week cutoff comparison. These were not externally preregistered. The announcement estimate provides no evidence of the predicted acceleration. The cutoff estimate has the predicted sign at both endpoints and under both notice restrictions. It nevertheless failed the original promotion gate: only eleven of twenty endpoint-specific window and donut estimates have wild-cluster $p < 0.05$; the upper-endpoint pseudo-cutoff rank is 0.143 rather than below 0.10; and 27 of 2,515 reconstructed dates (1.07 percent) change cells across endpoints, exceeding a denominator-specific .5 percent ambiguity threshold even though they are only 0.36 percent of all matched transactions. An earlier exact-date screen had already shown the local pattern, so subsequent reconstruction and notice restriction validate a known secondary signal.

Two later additions follow the same discipline. First, a conditional-outcome protocol frozen on August 27, 2026, before inspection, fixed three outcomes on the cutoff cells (log one-plus marketing duration, hedonic log transacted unit price, and the log list-to-transacted ratio), their predicted signs, and restricted-null wild-school inference. Table A20 below re-evaluates those unchanged specifications on the headline notice-verified frame so that the flow and conditional panels share one sample. On the original sixty-one-community frame, the lower- and upper-date estimates were +0.733 and +0.775 for duration, −0.011 and −0.015 for price, and −0.012 at both endpoints for the price ratio. Second, a geographic displacement diagnostic was pre-specified on August 29, 2026 with predictions and all design choices (nearest-exposed distance, the below-median split, and the bizcircle adjacency notion) fixed from geometry alone before any outcome was estimated; all four registered tests are reported regardless of outcome.

The raw 2019 cells are essential for interpreting that sign: completed counts are approximately flat in exposed communities and rise sharply in ordinary-school communities. The cutoff coefficient is therefore a relative DDD shortfall, not an observed collapse in exposed transactions or a districtwide volume estimate. Table A7 reports the full post-protocol geographic anatomy. The four rows are mutually exclusive archive segments; the broader rows use raw totals and are neither assignment-verified controls nor untreated-market estimates. The observed rise is descriptive, not a predicted rebound or an identifying assumption. The absence of a broad Haidian grace-period response also differs from Xicheng, where Shao et al. (2023) report an announcement-period increase followed by an enforcement-period decline in transaction volume.

Table A7: Geographic anatomy of the winter-to-spring rise

	Lower completion date			Upper completion date		
Market	2018 A/B	2019 A/B	Ratio/ratio	2018 A/B	2019 A/B	Ratio/ratio
Formerly key	1.481	0.959	0.647	1.642	1.079	0.657
Ordinary	1.088	1.671	1.535	1.260	1.792	1.422
Other Haidian	1.361	1.199	0.881	1.495	1.332	0.891
Beijing outside Haidian	1.358	1.293	0.953	1.491	1.474	0.988

Notes: A/B is the after-to-before ratio in fixed sixteen-week windows around January 1. Ratio/ratio divides the 2019 A/B ratio by the identical 2018 calendar ratio. Formerly key and ordinary rows use inverse-reuse weighted mean transactions per community on the complete 59-community grid. Other Haidian (excluding that frame) and Beijing outside Haidian use raw archive totals. The groups are mutually exclusive segments from the same archive; broader groups are not assignment-verified controls. The table does not identify substitution or displaced transactions.

The main paper reports the preferred and literal-notice estimates with CR2 intervals and the exposed-school concentration. School sign counts are descriptive concentration diagnostics because controls are reused across matched sets.

The outcome-blind spatial match pairs all nineteen exposed communities to distinct notice-verified controls within 2.384 kilometers. School-equal post-cutoff effects are −0.718 and −0.728. Under a school-effect symmetry restriction, two-sided sign-flip sensitivities are 0.047 and 0.078; exact-notice controls produce 0.047 at both endpoints. These are not randomization-inference tests. Six of seven school effects are negative, every exposed-school omission preserves the negative sign, and symmetric Lunar-New-Year exclusions retain 78–96 percent of the corresponding magnitude. The 1.5 km rematch retains only twelve communities and six schools and is near zero; the 2.0 km match retains seventeen, and full support begins at 2.5 km. Baseline housing attributes are balanced, but the 2017 completed-flow standardized difference is about 0.40. The match is therefore corroboration of a local counterfactual, not a second independently decisive design.

Table A8: Outcome-blind spatial-pair validation

	Lower completion date			Upper completion date			Support
Specification	Estimate	Exact p	CR1 p	Estimate	Exact p	CR1 p	Pairs/schools
Exact/alias, 1.0 km	−0.879	0.188	0.173	−1.065	0.125	0.089	7/5
Exact/alias, 1.5 km	0.040	0.938	0.926	−0.065	0.906	0.883	12/6
Exact/alias, 2.0 km	-0.929^{**}	0.047	0.066	-0.813^{**}	0.047	0.077	17/7
Exact/alias, 2.5 km	-0.718^{**}	0.047	0.108	-0.728^{*}	0.078	0.116	19/7
Exact/alias, 3.0 km (primary)	-0.718^{**}	0.047	0.108	-0.728^{*}	0.078	0.116	19/7
Exact-notice controls, 3.0 km	-0.836^{**}	0.047	0.077	-0.845^{**}	0.047	0.078	19/7
Nearest with replacement, 3.0 km	-0.871^{**}	0.016	0.045	-0.752^{*}	0.094	0.121	19/7
LNY blackout ±14 days	−0.564	0.141	0.150	−0.565	0.172	0.178	19/7
LNY blackout ±21 days	−0.630	0.203	0.184	−0.643	0.219	0.225	19/7
LNY blackout ±28 days	−0.692	0.203	0.176	−0.693	0.188	0.223	19/7
Pre-policy placebo: 2018 versus 2017	0.459	0.281	0.323	0.470	0.344	0.365	19/7

Notes: Negative estimates use the paper's convention and denote an exposed-relative completed-flow shortfall. The match is maximum-cardinality, minimum-total-distance and uses only coordinates and the independent 2017 notice audit; except where stated, controls are distinct. The primary 3 km assignment was frozen before matched outcomes were opened. Estimates first average community-pair DDDs within exposed school and then average schools. Exact two-sided p-values enumerate all 2^G joint sign flips at the exposed-school level; stars use these values: $^{***}p < .01$, $^{**}p < .05$, $^{*}p < .10$. CR1 p-values use $t(G-1)$. Each caliper is rematched, so changing support explains why the 1.5 km row is not comparable to a simple trimming exercise. LNY rows symmetrically remove the indicated days around the year-specific Lunar New Year. The placebo compares the January 2018 differential with January 2017.

Table A9: Balance in the frozen one-to-one spatial match

Predetermined variable	Standardized mean difference
Longitude	−0.244
Latitude	0.037
2017 asinh completed flow, lower date	0.404
2017 asinh completed flow, upper date	0.399
Median log unit price	0.048
Median floor area	0.032
Median construction year	0.117

Notes: Differences are formerly-key minus ordinary, standardized by the pooled dispersion in the nineteen frozen pairs. Housing attributes are available for eighteen pairs. Matching uses coordinates and assignment provenance, not outcomes or these balance statistics. The approximately 0.40 pre-policy-flow difference motivates the later preperiod-balancing diagnostic and remains a limitation of the spatial comparison.

The full-sample surface exercise permits the experiment-side movement to vary smoothly across Haidian. Linear surfaces leave the coefficients slightly larger in magnitude and below 0.05 under school-wild inference at both endpoints. Quadratic surfaces preserve the magnitudes but widen school-wild p-values to 0.105 and 0.138. Conley standard errors are reported only as residual-dependence diagnostics: residual Moran statistics are close to zero and the Conley standard errors fall as the distance cutoff expands, so they do not supply independent identifying variation.

Table A10: Spatial robustness of the completed-flow triple difference

Specification	Coefficient	School-wild p	Conley p: 1/2/3 km
Panel A. Lower completion date, full preferred sample			
No spatial surface	-0.918^{**} (0.335)	0.012	0.015/0.007/0.003
Linear geography × cells	-0.972^{**} (0.372)	0.021	0.021/0.009/0.004
Quadratic geography × cells	-0.905 (0.432)	0.105	0.026/0.005/0.001
Panel B. Upper completion date, full preferred sample			
No spatial surface	-0.762^{**} (0.376)	0.035	0.057/0.039/0.030
Linear geography × cells	-0.839^{**} (0.395)	0.049	0.053/0.030/0.021
Quadratic geography × cells	-0.766 (0.448)	0.138	0.070/0.031/0.018
Panel C. Previously locked outcome-blind 3 km spatial match			
Lower completion date	-0.718^{**} (0.380)	0.047	–
Upper completion date	-0.728^{*} (0.397)	0.078	–

Notes: The outcome is the inverse-hyperbolic-sine of completed transactions per community in a 16-week cell. Every coefficient uses the paper's After × 2019 × Formerly-key convention, so a negative value is an exposed-relative, seasonally netted shortfall. Panels A and B retain all 59 communities and the original inverse-reuse weights. All models contain community fixed effects, experiment-by-side fixed effects, and lower-order exposure interactions. Linear and quadratic models permit the stated coordinate surface to differ in every experiment-side cell. Parentheses contain assigned-school CR1 standard errors; stars and the adjacent p-values use 9,999-draw restricted-null school-wild inference. Conley columns report two-sided $t(N-K)$ p-values for Bartlett kernels at 1, 2, and 3 km. Panel C reproduces the previously locked school-equal spatial-pair benchmark; its school-wild column instead reports the exact seven-school sign-flip value. $^{***}p < .01$, $^{**}p < .05$, $^{*}p < .10$.

A post-baseline balancing exercise uses the complete local donor universe rather than one control per exposed community. Nonnegative weights are chosen only from January 2015–August 2016 monthly completed flows and coordinates; a 2015 fit and January–August 2016 validation select the penalty before any later outcome enters the final weights. The preferred weights retain all forty controls, have effective sample sizes of 29.6 and 29.8, and put at most 5.6 and 6.4 percent on one control. The coefficients remain negative in the paper's convention across the complete fixed-penalty family and every recomputed omission of any of the twenty-one represented schools.

The fully held-out 2018-versus-2017 placebo is nevertheless sizeable and opposite-signed, with wild $p = 0.068$ and 0.049. It is not a clean null. As the main text's benchmark discussion documents, the 2017–2018 sequence is consistent with a differential shock and recovery around Beijing's March 17, 2017 credit tightening, which coincides with the 2017 after-window and was plausibly most binding in the school-district segment; that timing does not establish the mechanism. The placebo is the oscillation viewed on its own, and the 2016 point estimate is near zero. Multiyear comparisons were added only after observing that result. Relative to the average 2017–2018 seasonal gap, the 2019 estimates remain -0.547 and -0.642; relative to 2017 alone, they are smaller and imprecise. Tables A11 and A12 preserve both the formal result and the adverse diagnostic.

For completeness, Table A13 collects the influence, spatial, and balancing exercises by design status. Figure A2 displays the complete bandwidth/donut family whose specifications were fixed before date reconstruction and all thirteen earlier pseudo-cutoffs. These exhibits are diagnostic rather than an independent research design.

The pre-specified nearby-control diagnostic and its far-control and shared-bizcircle sensitivities are

Table A11: Spatially local preperiod-balanced validation

	Lower completion date			Upper completion date		
Specification	Estimate	CR2 p	Wild p	Estimate	CR2 p	Wild p
CV-selected penalty	-0.853^{**}	0.034	0.015	-0.955^{***}	0.028	0.009
Fixed $\lambda = 0$	-0.873^{***}	0.049	0.006	-0.876^{**}	0.094	0.027
Fixed $\lambda = 0.01$	-0.825^{**}	0.045	0.013	-1.003^{***}	0.023	0.004
Fixed $\lambda = 0.1$	-0.853^{**}	0.034	0.015	-0.955^{***}	0.028	0.009
Fixed $\lambda = 1$	-0.877^{**}	0.021	0.011	-0.972^{***}	0.022	0.008
Fixed $\lambda = 10$	-0.881^{***}	0.018	0.009	-0.978^{***}	0.021	0.009
Earlier calendar experiments, CV-selected weights						
2018 versus 2017 (fully held out)	0.611^{*}	0.080	0.068	0.627^{**}	0.062	0.049
2017 versus 2016 (partly in sample)	−0.160	0.581	0.578	−0.056	0.840	0.831

Notes: Negative estimates use the paper's convention and denote an exposed-relative completed-flow shortfall. Control weights are nonnegative, sum to one, and use all notice-verified controls within 3 km of at least one exposed community. The preferred penalty is chosen using January–August 2016 prediction after fitting on 2015; final weights use only January 2015–August 2016 monthly flows and coordinates. CR2 uses coefficient-specific Satterthwaite degrees of freedom. Wild p-values use 9,999 restricted-null draws over every represented exposed and control school; stars use those values. The design was formalized after an exploratory pilot. The fully held-out placebo is opposite-signed and is not a clean pretrend pass. $^{***}p < .01$, $^{**}p < .05$, $^{*}p < .10$.

Table A12: Year-specific seasonal gaps and multiyear diagnostics

	Lower completion date			Upper completion date		
Year/contrast	Estimate	CR2 p	Wild p	Estimate	CR2 p	Wild p
Panel A. Formerly-key-minus-ordinary after-minus-before gap by year						
2016	0.015	–	–	0.005	–	–
2017	-0.145	–	–	-0.051	–	–
2018	0.466	–	–	0.576	–	–
2019	-0.387	–	–	-0.380	–	–
Panel B. 2019 relative to alternative seasonal benchmarks						
2019 minus 2018 (formal)	-0.853^{**}	0.034	0.015	-0.955^{***}	0.028	0.009
2019 minus 2017	−0.241	0.345	0.325	−0.329	0.229	0.194
2019 minus average of 2017–2018	-0.547^{**}	0.059	0.036	-0.642^{**}	0.046	0.017
2019 minus average of 2016–2018	-0.499^{**}	0.085	0.049	-0.557^{**}	0.068	0.034

Notes: Negative values use the paper's after-minus-before convention. Panel A reports the weighted formerly-key-minus-ordinary seasonal gap by year. Panel B reports 2019 relative to alternative benchmarks. Only 2019 minus 2018 belongs to the formal protocol. The remaining rows were added after observing the sizeable opposite-signed held-out placebo and must be read as post-protocol diagnostics. Weights remain fixed at the CV-selected preperiod values. Stars use all-school wild-cluster p-values: $^{***}p < .01$, $^{**}p < .05$, $^{*}p < .10$.

Table A13: Robustness and influence diagnostics

Diagnostic	Lower date	Upper date
Preferred notice-verified effect	-0.918^{**} [0.014]	-0.762^{**} [0.034]
Exact-notice-only effect	-0.870^{**} [0.021]	-0.715^{*} [0.055]
Unweighted asinh effect	-0.753^{**} [0.027]	-0.715^{**} [0.048]
Leave-one-community/school range	[-1.146, -0.750]	[-0.994, -0.595]
Negative matched-set effects	16/19	16/19
Negative exposed-school effects	7/7	6/7
Recorded window/donut range	[-0.987, -0.274]	[-0.924, -0.029]
Negative window/donut effects	10/10	10/10
Absolute rank of real cutoff among 14 dates	1 of 14	2 of 14
Full sample: linear spatial surface	-0.972^{**} [0.021]	-0.839^{**} [0.049]
Full sample: quadratic spatial surface	-0.905 [0.105]	-0.766 [0.138]
Spatial pairs: 3 km, exact/alias	-0.718^{**} [0.047]	-0.728^{*} [0.078]
Spatial pairs: 3 km, exact notice	-0.836^{**} [0.047]	-0.845^{**} [0.047]
Spatial pairs: pre-policy placebo	0.459 [0.281]	0.470 [0.344]
Local preperiod-balanced controls	-0.853^{**} [0.015]	-0.955^{***} [0.009]
Balanced held-out placebo: 2018 vs. 2017	0.611^{*} [0.068]	0.627^{**} [0.049]
Balanced 2019 vs. 2017–18 mean (post-protocol)	-0.547^{**} [0.036]	-0.642^{**} [0.017]

Notes: Effects use the main-text sign convention; negative values denote a lower formerly-key-minus-ordinary after-before contrast in 2019 than in the comparison experiment. They do not require formerly key-school counts to fall in levels. The preferred and exact-notice rows use exhaustive restricted-null wild inference over all 2^{21} assigned-school Rademacher sign vectors. Spatial-surface and balanced rows retain their recorded 9,999-draw assigned-school wild p-values; the unweighted row uses its assigned-school CR1 p-value. All original-design diagnostics preserve the original matching when units are omitted. The window/donut family contains seven symmetric windows and three 16-week donut exclusions for each timing endpoint. Rank 1 denotes the largest absolute coefficient among the real January 2019 cutoff and thirteen earlier pseudo-cutoffs. Linear and quadratic surfaces interact centered geography with every experiment-side cell. The separately frozen one-to-one spatial-pair rows use exact two-sided exposed-school sign-flip inference. Balanced-control weights are nonnegative and selected only with January 2015–August 2016 monthly flows and coordinates; their design was formalized after an exploratory pilot. The 2019-versus-2017–18 row was added after the opposite-signed held-out placebo was observed. Stars follow $^{***}p < .01$, $^{**}p < .05$, $^{*}p < .10$. School sign counts from the original design describe concentration and are not design-exact inference.

Table A14: Long-horizon persistence and falsification

	PPML relative-flow ratio		Asinh school-wild p-value	
Window and benchmark	Lower	Upper	Lower	Upper
16-week cutoff / identical 2018 calendar	0.422	0.462	0.012	0.035
Calendar 2019 / pooled 2015–2017	0.837	0.843	0.145	0.114
Calendar 2020 / pooled 2015–2017	1.021	1.013	0.871	0.893
Jan. 2019–Dec. 2020 / Jan. 2016–Dec. 2017	0.945	0.945	0.815	0.835
2017 placebo / pooled 2015–2016	0.881	0.887	0.121	0.193

Notes: Ratios below one denote lower formerly-key-relative completed flow. The first row reproduces the main sixteen-week design and exhausts all 2^{21} assigned-school Rademacher sign vectors under the restricted-null, fully studentized wild procedure. The remaining rows come from a separately specified persistence exercise estimated after the core cutoff analysis and retain 9,999 restricted-null draws clustered by historically assigned school. Annual models compare each indicated year with pooled pre-policy years; the two-year model compares cumulative calendar windows. The 2019 annual PPML cluster-t p-values are 0.063 and 0.064; the table reports ratios for scale rather than PPML significance. The similar 2017 placebo limits a causal interpretation of the calendar-2019 result.

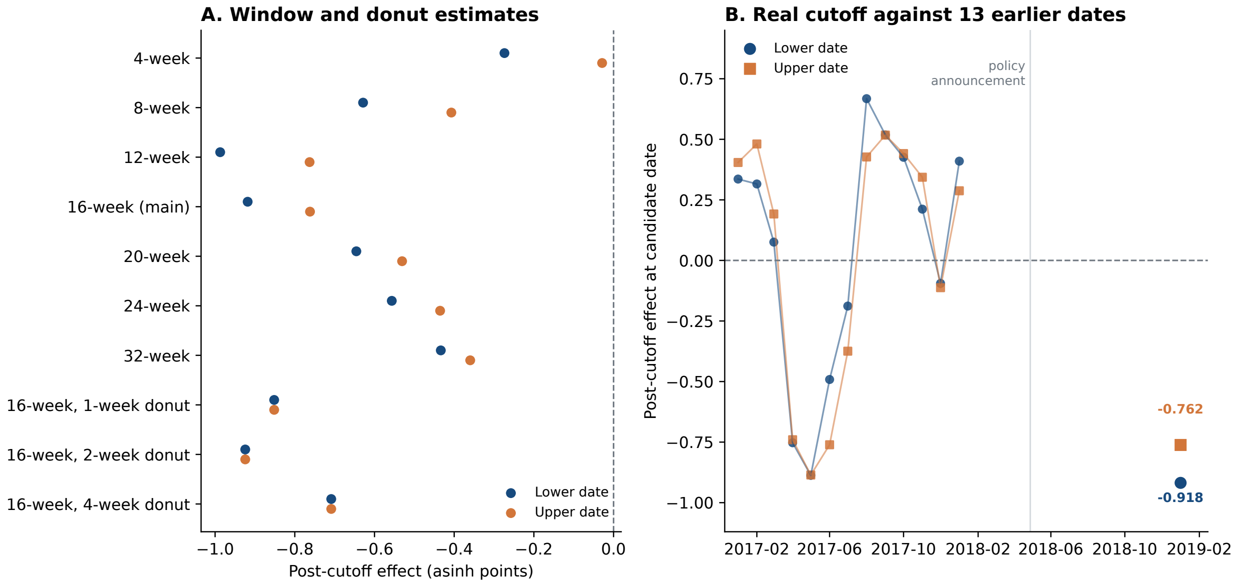

Figure A2: Specification and pseudo-cutoff diagnostics

Notes: Panel A displays every coefficient from the bandwidth/donut specification family fixed before date reconstruction. Panel B displays the real January 2019 coefficient and thirteen earlier pseudo-cutoffs. Coefficients are shown without significance filtering; preferred few-cluster inference appears in the main paper.

reported in Table A15.

Table A15: Geographic displacement diagnostics

	Lower date	Upper date
Near-control extra completed-flow increase (controls only)	−0.322 (0.426)	−0.040 (0.450)
wild-school p	0.393	0.918
Spearman correlation, control increase vs. distance	+0.068	−0.062
Headline DDD, far controls only	−1.003 (0.414)	−0.773 (0.465)
wild-school p; PPML ratio	0.021; 0.406	0.092; 0.467
Headline DDD, excluding same-bizcircle controls	−0.930 (0.441)	−0.703 (0.483)
wild-school p; PPML ratio	0.047; 0.417	0.155; 0.473

Notes: Pre-specified diagnostics for displacement of exposed-community transactions into nearby ordinary-school controls. Distance is from each control community to its nearest exposed community; the forty controls sit close to exposed communities (median nearest distance 1.00 km, maximum 2.54 km), so a substitution response should be visible in the near half. "Near" is below-median distance (20 of 40); the first row interacts the near indicator with the full experiment-side structure in the control-only cell panel, so a positive value would indicate an increase concentrated beside exposed communities. The middle row reports the rank correlation between each control community's own difference-in-differences and its distance. The lower rows re-estimate the headline design keeping only far controls, and excluding the 20 controls that share a bizcircle with an exposed community. Wild-school p-values use the restricted-null studentized procedure with 9,999 draws.

The matched pairs also permit a descriptive local decomposition. The formerly-key member's share of pair flow shifts by −10.4 and −11.5 percentage points, while combined pair flow shifts by −0.140 and −0.214 asinh points. Neither component is separately precise under the exhaustive seven-school sign-flip sensitivity, which requires school-effect symmetry and is not randomization inference. Because the nineteen pairs do not

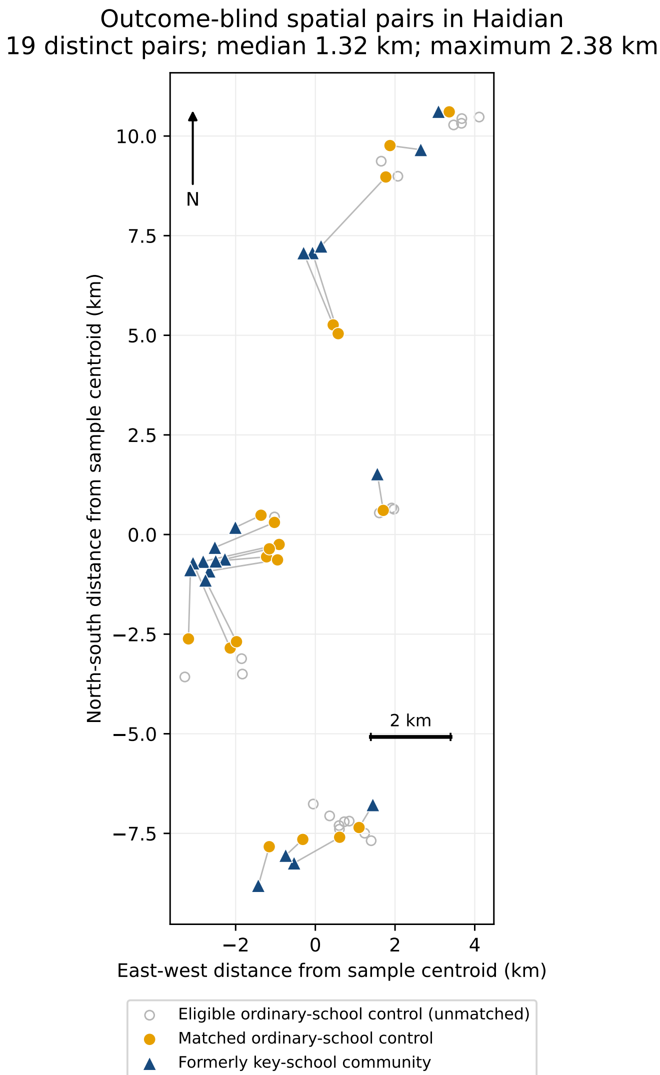

Figure A3: Outcome-blind one-to-one spatial pairs

Notes: Lines connect matched communities; they are not school-catchment boundaries. The global assignment maximizes retained exposed communities and then minimizes total geographic distance, without replacement, using only coordinates and the independent 2017 assignment audit. Coordinates share the same source coordinate system, so pair distances and relative positions are internally comparable.

form a closed market, this diagnostic cannot distinguish local diversion, flows to unmatched locations, and aggregate suppression.

Table A16: Local spatial-pair total and share diagnostics

	Lower completion date		Upper completion date	
Outcome	Coefficient	Exact p	Coefficient	Exact p
Asinh pair-total transactions	−0.140 (0.321)	0.688	−0.214 (0.343)	0.547
Formerly-key transaction share (p.p.)	−10.372 (6.915)	0.172	−11.532 (6.776)	0.125

Notes: The sample is the 19 distinct outcome-blind 3 km pairs. Each coefficient is the 2019 after-minus-before change net of the identical 2018 change, averaged first within treated school and then equally across the seven treated schools. Parentheses report CR1 treated-school standard errors; p-values and stars use the exact two-sided 2^7 school sign-flip distribution. A negative combined-total coefficient is a local pair-level shortfall. A negative share coefficient is a movement away from the formerly key member of the pair. The pairs do not form a closed market, so neither outcome identifies districtwide trade suppression. $^{***}p < .01$, $^{**}p < .05$, $^{*}p < .10$.

Table A17: Analysis hierarchy and falsification checks

Check	Lower timing endpoint	Upper timing endpoint	Interpretation
Notice-verified cutoff effect	−0.918** (wild p=0.014)	−0.762** (wild p=0.034)	Main completed-flow estimate
Exact-notice controls only	−0.870** (wild p=0.021)	−0.715* (wild p=0.055)	Assignment sensitivity
Monthly pre-period slope (61-community frame)	cluster-t p=0.029; wild p=0.121	cluster-t p=0.059; wild p=0.190	Inference differs with small-cluster correction
Real-cutoff rank among earlier dates	rank p=0.071	rank p=0.143	Only the lower endpoint falls below the 10 percent rank threshold
Announcement-to-cutoff acceleration	-0.380 (wild p=0.308)	-0.336 (wild p=0.392)	Estimate recorded before date repair

Notes: Cutoff effects use the main-table sign, so negative values denote a lower formerly-key-minus-ordinary after-before contrast in 2019 than in the identical 2018 calendar experiment; they do not require formerly key-school counts to decline in levels. Stars use restricted-null wild-cluster inference: $^{***}p < .01$, $^{**}p < .05$, $^{*}p < .10$. Lower and upper are the two endpoints of the fixed one-day reconstruction interval. Headline and exact-notice rows use the 59- and 50-community frames; rows explicitly labeled 61-community use the original broader frame. In the 414,979-transaction validation sample, all but one reconstructed date is within one day of the reported date. In the broader frame, 27 of 2,515 reconstructed dates (1.1 percent, or 0.36 percent of all 7,410 records) change cutoff-window membership across endpoints. All 20 endpoint-specific bandwidth/donut specifications were fixed before date reconstruction and later produced the predicted sign, but only 11 have wild-cluster $p < .05$. The cutoff was a secondary estimand in timestamped internal protocols, not an external preregistration.

D Completed-Pipeline and Public-Stock Accounting

Let $I_i = 1$ denote that listing i appears as a completed transaction before the archive ends in January 2021. Define selected end-of-day active stock, posting, and exit as

$$S^C_{ct} = \sum_{i:c_i=c} I_i \mathbf{1}\{a_i \le t < d_i\}, \tag{A7}$$

$$L^C_{ct} = \sum_{i:c_i=c} I_i \mathbf{1}\{a_i = t\}, \tag{A8}$$

$$X_{ct} = \sum_{i:c_i=c} I_i \mathbf{1}\{d_i = t\}. \tag{A9}$$

They obey $S^C_{ct} = S^C_{c,t-1} + L^C_{ct} - X_{ct}$. The implemented identity has zero error in every community-period cell under both date conventions. This stock conditions on completion by the archive end and is not the full offered inventory.

Table A18: Selected observed-completer pipeline accounting

	Lower date	Upper date
Selected-posting PPML ratio	0.962	0.962
Completed-exit PPML ratio	0.422	0.462
Maximum accounting-identity error	0	0

Notes: Posting, stock, and exit are defined only within listings observed to complete by the January 2021 archive end. End-of-day stock obeys $\Delta S^C = L^C - X^C$ exactly in every community–experiment–side cell. Ratios below one denote a weaker formerly-key-relative 2019 after response after subtracting the identical 2018 calendar contrast. These objects do not measure full offered inventory, withdrawal, relisting, later completion, or a marketwide sale hazard.

D.1 Completion timing within fixed posting cohorts

A post-protocol diagnostic fixes posting cohorts from January 1 through April 22 in 2018 and in the identical 2019 calendar window, then asks whether listings observed to complete by the archive end do so by each of seven horizons. Both cohorts receive at least 540 days of follow-up. Table A19 reports every horizon. At thirty days, the formerly-key-relative 2019 shift is −0.280 and −0.277 under the lower and upper completion dates; CR2 and restricted-null school-wild p-values are approximately 0.04. No horizon survives Holm adjustment within either completion-date endpoint. The 2019 treated cell contains 155 listings from seventeen communities but only five treated schools. Because the sample conditions on observed completion, long-horizon convergence is mechanical and does not imply that all offered homes eventually sell.

Table A19: Clearance within fixed posting cohorts of observed completers

	Community-FE DID		CR2 inference		Wild inference	
Days	Lower	Upper	p_L	p_U	p_L	p_U
30	-0.280	-0.277	0.040	0.041	0.037	0.036
60	-0.281	-0.277	0.205	0.215	0.175	0.190
90	-0.202	-0.202	0.342	0.342	0.407	0.412
112	-0.154	-0.167	0.405	0.396	0.469	0.468
180	-0.046	-0.046	0.704	0.704	0.730	0.731
365	-0.063	-0.063	0.222	0.222	0.244	0.245
540	+0.007	+0.007	0.801	0.801	0.797	0.794

Notes: The sample contains 709 listings posted from January 1 through April 22 in 2018 or 2019 and observed to complete by the January 2021 archive end; both cohorts have at least 540 days of follow-up. The 2019 treated cell has 155 listings from 17 communities but only 5 treated schools. Each coefficient is the formerly-key versus ordinary change in completion-by-horizon probability between the 2018 and 2019 posting cohorts, with community and cohort-year fixed effects. CR2 uses coefficient-specific Satterthwaite degrees of freedom; wild values use 9,999 restricted-null assigned-school draws. At day 30, Holm-adjusted p-values are 0.279 and 0.290 under CR2 and 0.261 and 0.256 under the wild procedure for the lower and upper dates; no horizon survives adjustment across seven horizons. Conditioning on observed completion excludes withdrawals and unresolved or later-completing listings. Long-horizon convergence is mechanical within this selected sample, not evidence that all offered homes eventually sell.

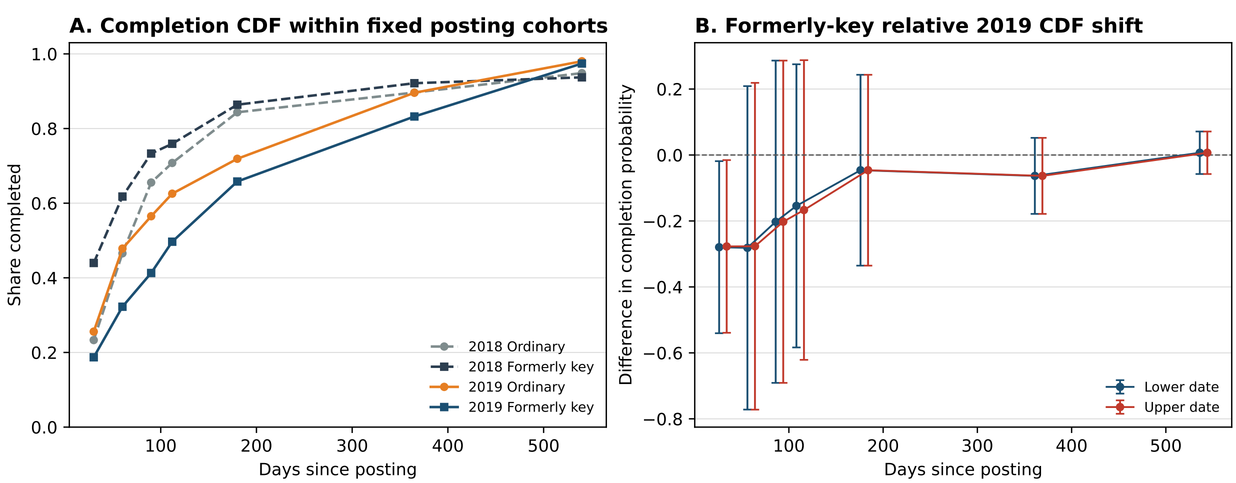

Figure A4: Among observed completers, early completion shifts are negative in point estimates

Notes: Posting cohorts run from January 1 through April 22 in 2018 and 2019, receive at least 540 days of follow-up, and condition on completion by the archive end. Panel A reports design-weighted raw CDFs at the lower completion date. Panel B reports community-fixed-effect difference-in-differences estimates with CR2/Satterthwaite 95 percent intervals under both completion-date conventions. The day-30 unadjusted p-values are approximately 0.04, but no horizon survives Holm adjustment across seven horizons. The 2019 treated cell has 155 listings from seventeen communities and five treated schools. Long-horizon convergence is mechanical in this selected sample, not evidence that all offered homes eventually sell.

D.2 Duration and prices conditional on completed sales

Table A20 reports the complete pre-specified conditional-outcome family on the headline notice-verified frame. Every row conditions on a completed transaction. The duration estimate therefore describes the selected distribution of successful spells, while the price and discount estimates describe selection among trades that clear; none is a full-market sale probability or duration distribution.

Table A20: Incidence around the cutoff: trade, time, and the price of what trades

	Lower date	Upper date
Panel A. Completed transaction flow (community cells, n = 236)		
Asinh triple difference	−0.918 (0.335)	−0.762 (0.376)
exhaustive wild-school p	0.012	0.035
PPML completed-flow ratio	0.422	0.462
Panel B. Conditional on a completed sale (transactions, n = 1,173)		
Log(1 + marketing duration)	+0.699 (0.422)	+0.745 (0.274)
wild-school p [Holm]	0.238 [0.714]	0.043 [0.130]
Log transacted price per m^2	−0.011 (0.043)	−0.016 (0.041)
wild-school p [Holm]	0.795 [0.856]	0.686 [1.000]
Log list-price-to-transacted-price ratio	−0.013 (0.016)	−0.012 (0.018)
wild-school p [Holm]	0.428 [0.856]	0.534 [1.000]

Notes: All coefficients are exposed-relative after-minus-before effects net of the 2018 seasonal contrast, in the notice-verified preferred frame with headline design weights and assigned-school clustering. Panel A repeats the paper's headline completed-flow design. Panel B applies the pre-specified conditional-outcome models to the same frame: community and experiment-side fixed effects with lower-order exposure interactions; the price and list-to-transacted regressions add log size, construction year, and floor-plan controls. Wild-school p-values use the restricted-null studentized procedure with 9,999 draws; brackets apply a Holm correction across the three Panel B outcomes. Panel B conditions on sale, so these are selected outcomes: under the framework the transacted-price response is not signed, and its null is not evidence about capitalization.

Table A21 reports the full outcome family fixed before the attention coefficients were inspected, together with the subsequent scale-corrected physical-showing-rate audit. The duration component passes its predeclared sign and inference criterion, but the joint mechanism-promotion gate fails because corrected daily-rate noninferiority does not. Cumulative showings are consequently treated as exposure-loaded description rather than evidence of intact buyer arrival.

D.3 A frozen selected-duration tail diagnostic

The mean selected-duration result motivates one distributional diagnostic. Before estimating its policy coefficient, I fixed the long-duration threshold at 85 days, the nearest-rank 90th percentile among preferred-frame completed transactions with finite, nonnegative durations whose lower feasible completion date fell in 2015–2016; fixed the same weighted DDD, completion-date endpoints, school clustering, wild and CR2 inference; and required the predicted sign to survive every treated-school omission at both endpoints. Table A22 reports the complete prespecified estimate and inference family. The after-effect is positive at both endpoints and under all fourteen school-by-endpoint omissions, but the confidence intervals are wide. This diagnostic therefore locates the selected response in the right tail without establishing a marketwide duration distribution or a precise tail effect.

Table A21: Complete selected-completer outcome family

Panel A. Entry and exit in the completed-listing pipeline

	Lower date	Upper date
Selected-entry PPML ratio	0.962	0.962
Completed-exit PPML ratio	0.422	0.462
Posting events / completed exits	1,364 / 1,173	
Community–experiment–side cells	236	

Panel B. Frozen exact-linked attention and duration outcomes

Outcome, log-one-plus scale	DDD	CR2 p	Wild p	N / clusters
Physical showings per day	−0.069	0.432	0.402	855 / 19
Followers per day	0.064	0.584	0.551	855 / 19
Page views per day	0.148	0.739	0.716	755 / 17
Cumulative physical showings	0.854	0.064	0.040	855 / 19
Marketing duration	0.857	0.035	0.019	855 / 19

Panel C. Scale-corrected physical-showing rate audit

Selected-completer PPML rate ratio	0.728
Two-sided school-clustered p-value against one	0.497
One-sided 95 percent lower rate-ratio bound	0.328
Wild one-sided p-value for H_0 : RR $\leq$ 0.75	0.533
Predeclared 75 percent noninferiority gate	Fails

Notes: Every panel conditions on completion observed by the archive end and therefore does not measure the offered-listing universe. Panel B reports the complete frozen outcome family; coefficients are exposed-relative post effects after subtracting the identical 2018 calendar contrast. Wild tests use 9,999 restricted-null draws. The 855-record support contains 18 exposed and 38 control communities assigned to 6 exposed and 13 control schools; the page-view outcome has narrower support. Cumulative showings can rise mechanically with duration. Panel C uses a count model with listing days as exposure and supersedes an invalid noninferiority interpretation of the transformed showings-per-day outcome.

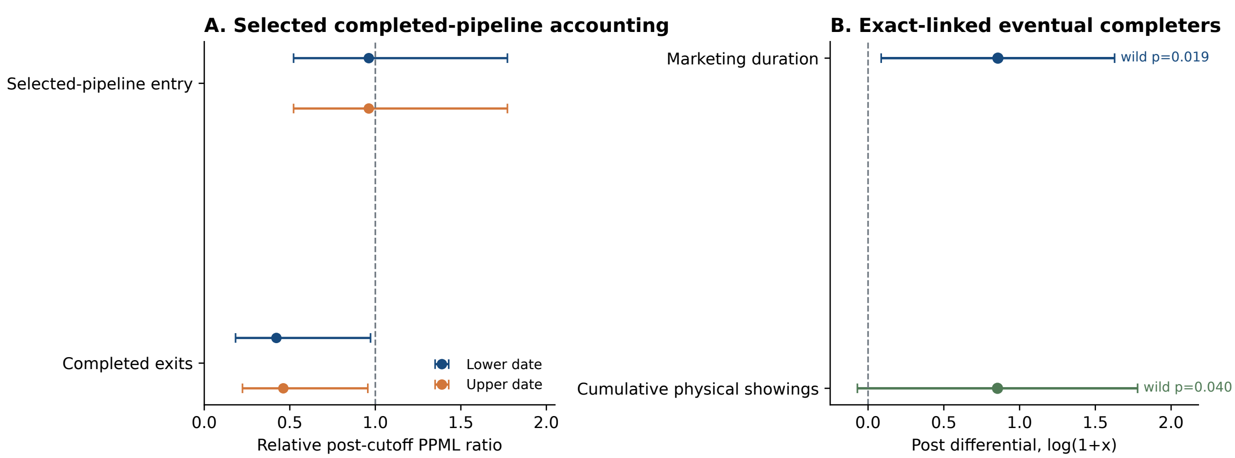

Figure A5: Timing and attention inside the observed-completer pipeline

Notes: Panel A reports exposed-relative PPML ratios for posting into, and exit from, the set of listings observed to complete by the archive end. Panel B reports real-versus-identical-calendar DDD estimates for the unique exact-linked observed-completer sample. Intervals use the displayed equation-specific inference. Neither panel describes the full offered-listing universe.

Table A22: Selected completed trades shift toward the long-duration tail

	Lower completion date	Upper completion date
After-minus-before DDD (percentage points)	10.2	18.1
CR1S standard error (percentage points)	12.9	9.7
CR1S p-value	0.437	0.077
CR2 standard error (percentage points)	14.7	10.6
CR2 95% interval (percentage points)	[−27.0, 47.4]	[−8.7, 44.9]
School-wild p-value	0.572	0.136
CR2/Satterthwaite p-value	0.516	0.145
Treated-school leave-out range (percentage points)	[8.6, 12.3]	[15.8, 22.6]
Observations	1,173	1,173
Assigned-school clusters	20	20

Notes: Before estimating the policy coefficient, the long-duration threshold was frozen at 85 days, the nearest-rank 90th percentile among 2,171 preferred-frame completed transactions with finite, nonnegative durations whose lower feasible completion date fell in 2015–2016. The outcome is one when recorded marketing duration is at least 85 days. Entries report the after-minus-before form of the weighted seasonal DDD; positive values indicate a larger exposed-relative long-tail share after implementation. CR1S denotes the conventional finite-cluster-corrected school-cluster covariance. The school-wild procedure uses 9,999 restricted-null draws. CR2 uses an identity working model with Satterthwaite degrees of freedom of 5.31 and 5.28. All seven treated-school omissions preserve the positive after-effect at both date endpoints. The sample conditions on completion and does not estimate the duration distribution of all offered listings.

The offered-stock comparison uses three independent public cross-sections: a November 28–29, 2017 crawl with exact listing IDs and a source-native community index (Huang, 2017); the H4M real-estate cross-section, whose contents are consistent with a late-March 2018 active-listing wave (Zhao et al., 2022); and a broad February 27, 2019 community-by-community crawl with 41,290 active rows across 4,131 named communities. Source-code review confirms iteration over every detected community page and every detected listing page, and the wave has positive rows for 54 of the 59 frozen communities. It nevertheless omits listing and dwelling IDs, row timestamps, and a request-success ledger, so it is not a certified universe. The union of the three waves is not a listing panel and does not reveal individual entry, withdrawal, relisting, or terminal states.

For the full offered universe, let S^O_{ct} be end-of-period active stock, E^O_{ct} the period's pre-exit exposure to sale, L^O_{ct} new entries, W^O_{ct} withdrawals, and h_{ct} the execution rate. The accounting system is

$$E^O_{ct} = S^O_{c,t-1} + L^O_{ct}, \tag{A10}$$

$$X_{ct} = E^O_{ct} h_{ct}, \tag{A11}$$

$$S^O_{ct} = E^O_{ct} - X_{ct} - W^O_{ct}. \tag{A12}$$

For aligned exposed-comparison period ratios, this implies $R_F = R_E R_h$, where R_F is relative completed flow, R_E relative period exposure, and R_h the relative execution rate. For baseline source $j \in \{\text{November 2017, H4 March 2018}\}$, the plotted cross-wave stock point is

$$\widetilde{R}_{S,j} = \frac{(S_K/S_O)_{\text{February 2019},j}}{(S_K/S_O)_{\text{baseline},j}}, \tag{A13}$$

computed on the identical community support available in source j.

The public snapshots measure a point-stock contrast $\widetilde{R}_S$, not period exposure R_E. Define $\kappa \equiv \widetilde{R}_S/R_E > 0$ to collect treatment-differential crawler coverage, the unobserved cross-year pre-period stock ratio, and point-snapshot-to-period-exposure alignment. Then

$$R_h = \frac{R_F \kappa}{\widetilde{R}_S}. \tag{A14}$$

Proposition A3 (Comparability-indexed point calibration)**.** *If an analyst posits that the combined comparability multiplier belongs to $K = [\underline{\kappa}, \overline{\kappa}]$, then, conditional on point values $(R_F, \widetilde{R}_S)$,*

$$\mathcal{H}(K) = \left[\frac{R_F \underline{\kappa}}{\widetilde{R}_S}, \frac{R_F \overline{\kappa}}{\widetilde{R}_S}\right]. \tag{A15}$$

No execution change is compatible with the points if and only if $\widetilde{R}_S / R_F \in K$.

No substantively defended K is estimated from these crawls. The displayed set is therefore an assumption-indexed mapping of point estimates, not a confidence set or an estimated partial-identification region.

Table A23: Completed flow and public active stock on identical supports

Common support	Object	Relative ratio	95% interval	Communities
Nov. 2017–Feb. 2019 common support	Completed flow (lower date)	0.420	[0.167, 1.057]	54
Nov. 2017–Feb. 2019 common support	Completed flow (upper date)	0.464	[0.209, 1.033]	54
Nov. 2017–Feb. 2019 common support	Public active-stock contrast	0.830	[0.477, 1.447]	54
Late-Mar. 2018–Feb. 2019 H4 support	Completed flow (lower date)	0.368	[0.117, 1.154]	37
Late-Mar. 2018–Feb. 2019 H4 support	Completed flow (upper date)	0.427	[0.160, 1.142]	37
Late-Mar. 2018–Feb. 2019 H4 support	Public active-stock contrast	0.923	[0.557, 1.531]	37

Notes: Every stock-flow comparison uses identical community support within the named support. Completed-flow ratios compare the formerly-key/ordinary after-before ratio in 2019 with its identical-calendar 2018 analogue. Public active-stock rows instead compare exposed-relative point stocks between the two named crawler waves; they are not transaction-DDD estimands. Ratios below one therefore denote a lower exposed-relative completed-flow response or a lower cross-wave exposed-relative stock contrast, respectively; they do not imply a decline in exposed levels. Intervals use equation-specific school-clustered t critical values and are not joint confidence sets for the stock-flow gap. The public waves are separate crawler snapshots rather than a listing-transition panel. On the 54-community November support, the stock/flow point-ratio gaps are 1.97 and 1.79; joint school-score equality tests give $p = .085$ and $p = .137$ using the geometric mean of the equation-specific CR1 finite-sample corrections. The smaller H4 support gives larger point gaps but only four exposed-school clusters. This is cross-crawler comparability sensitivity, not an identified execution hazard.

On the November-source support, the exposed-relative stock point is 0.830 and the exposed-relative completed-flow ratios are 0.420 and 0.464. On the H4 support, the corresponding stock point is 0.923 and the completed-flow ratios are 0.368 and 0.427. Equality inference is correction-sensitive and neither source uniformly rejects a supply-only explanation under every finite-sample correction.

At $\kappa = 1$, the points imply execution-rate ratios of approximately 0.40–0.56. A supply-only point fit requires κ between 1.79 and 2.51. The comparison makes the required stock adjustment transparent but does not separately identify buyer arrival, match efficiency, acceptance, withdrawal, or bargaining.

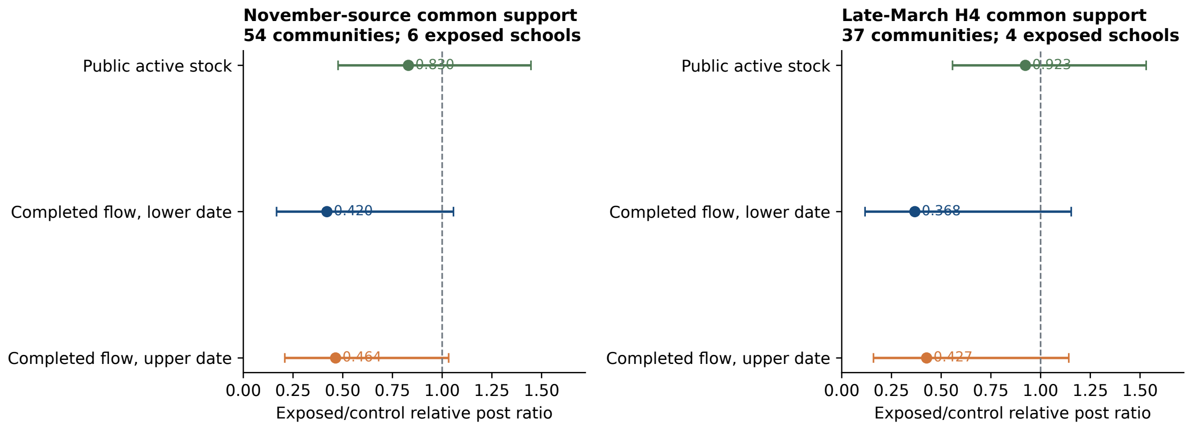

Figure A6: Completed flow and public active stock on identical supports

Notes: Intervals describe equation-specific uncertainty and do not form a joint confidence set for the stock-flow difference. Each comparison uses identical community support within data source.

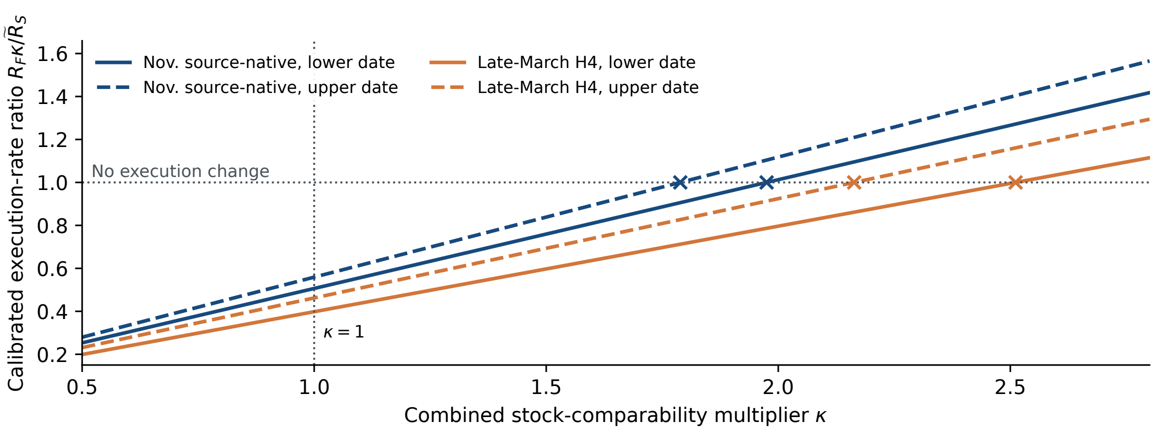

Figure A7: Comparability sensitivity for the execution interpretation

Notes: Curves apply the stock-flow identity to point estimates on identical supports. Crosses mark the comparability multiplier required for no execution change. The figure is not a confidence set and assigns no probability to κ.

E Fixed-Weight Relabeling and Unit-Type Diagnostics

This section reports two extensions governed by an internal analysis protocol written after the headline Haidian estimates were known and before these extensions were estimated. The protocol was not externally registered and should not be read as outcome-blind preregistration. It is archived with the code as `Replication/lift_2026-09-01/PRESPEC.md`.

The protocol also contained a proposed Dongcheng extension. I exclude that exercise from the evidence after primary-source verification invalidated its timing interpretation. An official Dongcheng admissions FAQ states that purchasers after June 30, 2018 who obtained a real-property certificate by December 31 retained single-school assignment; only certificates obtained after December 31 entered multi-school assignment. Completion dates around June 30 therefore do not implement a title-vintage reset. The code and outputs remain in the replication archive so that this design rejection is auditable; they are not pooled with Haidian or interpreted as a replication.

E.1 Exhaustive fixed-weight relabeling over assigned-school clusters

Table A24 re-estimates the headline design for every one of the $\binom{21}{7} = 116{,}280$ ways of designating seven of the twenty-one assigned-school clusters as exposed, holding the frozen panel, design weights, and fixed effects fixed. Ranked from the most negative coefficient, the actual grouping is 276th and 1,743rd at the two completion-date conventions; ranked by absolute magnitude, it is 2,989th and 5,656th. Descriptive two-sided tail shares are 0.026 and 0.049 on the coefficient and 0.011 and 0.042 on the assigned-school CR1 t-statistic. The 95th percentiles of the absolute placebo coefficients are 0.816 and 0.757, against absolute observed effects of 0.918 and 0.762; the upper-date comparison is consequently close to that threshold. Key-school status was not assigned, the frame and inverse-reuse weights were constructed around the actual exposure map, and those weights remain fixed under relabeling. The exercise is therefore a finite label-sensitivity description, not design-based or randomization inference and not an independent confirmation of the wild bootstrap.

Table A24: Exhaustive fixed-weight relabeling over assigned-school clusters

	Lower date	Upper date
Observed after-minus-before DDD	−0.918	−0.762
Label assignments (7 of 21 schools, all subsets)	116,280	116,280
Signed rank (most negative = 1)	276	1743
Absolute rank (largest $\lvert DDD^{*}\rvert$ = 1)	2989	5656
Two-sided tail share, coefficient	0.026	0.049
Two-sided tail share, studentized (CR1)	0.011	0.042
Lower-tail share, coefficient	0.002	0.015
95th percentile of $\lvert DDD^{*}\rvert$	0.816	0.757

Notes: For every one of the $\binom{21}{7}$ ways of designating seven of the twenty-one assigned-school clusters as exposed, all communities in those schools are relabeled and the production weighted-asinh DDD is re-estimated with the frozen frame, inverse-reuse weights, community fixed effects, and experiment-side fixed effects. Tail shares describe the finite set of relabelings. Key-school status was not assigned, and the frame and weights were constructed around the actual exposure map and held fixed under relabeling. The reported shares are therefore label-sensitivity summaries, not randomization p-values or an independent inference procedure.

E.2 Unit-type split

The unit-type split reported in the main text recorded two competing readings before estimation: a larger gap in family-sized units, or a larger gap in small units bought mainly for the school right. The 60 square meter cutoff was fixed as a round threshold near the 2019 window's first quartile; it classifies 262 of the 1,173 observations in the four principal cells as small. The bedroom split uses the archive's room-count field and omits seven observations with missing room counts. Class-specific estimates re-run the production weighted-asinh design within class; the class difference comes from a pooled model with class-specific fixed effects. The floor-area split satisfies the recorded class-difference and PPML-ordering rule under the designated 2018 benchmark, although its interaction p-values are 0.050–0.052 and older benchmarks attenuate the contrast. The bedroom split does not satisfy the PPML-ordering rule: its PPML ratios are slightly lower for one-bedroom than for two-plus-bedroom units. The evidence is therefore a benchmark-specific heterogeneity diagnostic, not a buyer-family mechanism test.

Table A25 reports the raw 2019 class differences and identical-calendar comparisons against 2016, 2017, and 2018. These comparisons were constructed after inspecting the designated-benchmark heterogeneity. The raw 2019 floor-area interactions are small and imprecise, and none of the floor-area or bedroom interactions survives adjustment across the four benchmark comparisons within an endpoint. This post-estimation sensitivity table is reported to show exactly where the designated split obtains its apparent strength.

Table A25: Unit-type sensitivity across seasonal benchmarks

Comparison	Small-unit estimate	Large-unit estimate	Small minus large	Wild p	Holm p
Panel A. Floor area (≤ 60 m^2 versus > 60 m^2)					
Lower completion date					
Raw 2019 DD	−0.333 (0.235)	−0.457 (0.215)	+0.124 (0.185)	0.437	1.000
2019 minus 2016 DDD	−0.048 (0.330)	−0.181 (0.373)	+0.133 (0.373)	0.682	1.000
2019 minus 2017 DDD	−0.225 (0.260)	−0.508 (0.264)	+0.283 (0.314)	0.345	1.000
2019 minus 2018 DDD (designated)	−0.237 (0.330)	−0.963 (0.318)	+0.726 (0.343)	0.052	0.209
Upper completion date					
Raw 2019 DD	−0.339 (0.234)	−0.339 (0.213)	−0.000 (0.159)	0.998	1.000
2019 minus 2016 DDD	−0.054 (0.331)	−0.063 (0.380)	+0.009 (0.398)	0.979	1.000
2019 minus 2017 DDD	−0.231 (0.259)	−0.472 (0.264)	+0.241 (0.340)	0.438	1.000
2019 minus 2018 DDD (designated)	−0.152 (0.373)	−0.904 (0.333)	+0.752 (0.384)	0.050	0.200
Panel B. Bedrooms (one versus two or more)					
Lower completion date					
Raw 2019 DD	−0.165 (0.131)	−0.412 (0.262)	+0.247 (0.252)	0.233	0.466
2019 minus 2016 DDD	+0.227 (0.273)	−0.141 (0.434)	+0.368 (0.490)	0.459	0.466
2019 minus 2017 DDD	+0.037 (0.195)	−0.561 (0.261)	+0.597 (0.257)	0.034	0.131
2019 minus 2018 DDD (designated)	−0.180 (0.261)	−0.850 (0.297)	+0.669 (0.299)	0.033	0.131
Upper completion date					
Raw 2019 DD	−0.170 (0.131)	−0.299 (0.256)	+0.130 (0.257)	0.482	0.964
2019 minus 2016 DDD	+0.222 (0.274)	−0.028 (0.437)	+0.250 (0.499)	0.623	0.964
2019 minus 2017 DDD	+0.032 (0.195)	−0.531 (0.252)	+0.563 (0.282)	0.036	0.143
2019 minus 2018 DDD (designated)	−0.189 (0.281)	−0.786 (0.313)	+0.596 (0.335)	0.048	0.143

Notes: This post-estimation diagnostic uses the frozen 59-community frame, design weights, 16-week windows, and completion-date conventions. The raw 2019 row is the formerly-key-minus-ordinary after-before difference. Each DDD row subtracts the identical calendar contrast ending in the stated benchmark year. Estimates are weighted inverse-hyperbolic-sine coefficients; CR1 assigned-school standard errors are in parentheses. "Small minus large" is the pooled class interaction, so a positive value means a more negative gap for the large class. Wild-school p-values use 9,999 restricted-null studentized draws. Holm values adjust the four interaction tests within each split and completion-date endpoint. These comparisons were constructed after inspecting the designated-benchmark heterogeneity and are sensitivity diagnostics, not pre-specified tests.

F Reproducibility and Scope of Claims

Hashed inputs, assignment samples, and machine-readable result files are recorded in the analysis protocols. A core exhibit builder regenerates the underlying cutoff estimates and figures, while separate locked routines generate assignment-stability, road-network, persistence, pre-announcement value, and selected-attention audits. The submission tables are assembled from machine-readable outputs produced by these documented modules. Independent routines reproduce sample construction, reuse weights, regressions, wild-cluster and CR2 inference, aggregation, and leave-one-out estimates. For the four headline and exact-notice asinh coefficients, a standalone audit reproduces the archived 9,999-draw Monte Carlo counts and then exhausts all 2^{21} assigned-school Rademacher sign vectors under the same restricted-null, fully studentized procedure. The reported exhaustive values are exact for that bootstrap sign distribution, not assignment-based randomization p-values.

The replication folder accompanying this appendix contains the full assignment audit, frozen matched frame and pairs, machine-readable headline and long-horizon results, selected-completer summaries, the frozen selected-duration-tail protocol and script, and the incidence-promotion and displacement-diagnostic scripts with their machine-readable results. The main transaction input is proprietary and cannot be redistributed without provider permission. The public notice images, source URLs, and their hashes remain separately auditable.

The interpretation used throughout the paper is:

1. **Identified under parallel differential seasonality:** a cutoff-local exposed-versus-ordinary DDD in completed transaction flow, not an absolute exposed decline or districtwide contraction.
2. **Exactly accounted but selected:** entry, exit, and stock among eventual sales.
3. **Conditionally calibrated:** execution as a function of public stock and κ.
4. **Not identified:** full inventory, listing-level sale probability, withdrawal, relisting, buyer arrival, permanent trade destruction, or welfare.

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